

 QUANTITATIVE FINANCE

MASTER PROMPT GUIDE

AI-Powered Models for NIFTY 50, S&P 500 & Global Markets

Monte Carlo · Black-Scholes · Market Timing · ML Direction · Macro Prediction
Mean Reversion · Pairs Trading · Risk Factor · VaR · Portfolio Optimization

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Version 2.0 · March 2026 · 10 Models · 50+ Ready-to-Use Prompts

10 50+

MODELS PROMPTS

3

DIFFICULTY LEVELS

FREE

TO USE

SECTION 01 · ABOUT THIS GUIDE

This is the most comprehensive AI Prompt Guide ever created for Quantitative Finance — specifically built for Indian markets (NIFTY 50, NSE) and global instruments (S&P 500, Crypto, Forex). It compiles every model built by @deepthinksfinance into one master reference, and adds 5 brand-new models with full prompt guides.

You do NOT need to be a programmer. Simply copy any prompt and paste it into Claude AI, ChatGPT, or Gemini. The AI will generate complete, working Python code that runs on Google Colab for free.

How to Use This Guide

 Step-by-Step Process	 Customization Tips
Copy any prompt from this guide	Swap [BRACKETS] with your values
Paste into Claude / ChatGPT / Gemini	Customize instrument, timeframe, params
Copy the generated Python code	Add your own data or use auto-download
Paste into Google Colab	Share your notebook via Colab link
Press Run (▶) on each cell	Learn from the code comments
View your interactive charts	Post your results — tag @deepthinksfinance!

SECTION 02 · ALL 10 QUANT MODELS AT A GLANCE

All 10 models below are fully buildable using AI prompts — no manual coding required. Each model is designed for NIFTY 50 but can be adapted to any global instrument.



Monte Carlo Simulation Engine PROBABILISTIC

Simulates thousands of possible price paths using Geometric Brownian Motion (GBM). Outputs probability distributions, confidence bands, VaR, and 3D path visualizations.

20 prompts · GBM · VaR · 3D Plotly · NIFTY/S&P/Crypto



Black-Scholes Options Pricing Lab DERIVATIVES

Live NIFTY options pricing engine with all 5 Greeks (Delta, Gamma, Theta, Vega, Rho). Includes strategy builder with 8 strategies and interactive 3D payoff surfaces.

15 prompts · Greeks · Vol Smile · Strategy Builder · NSE Weekly F&O



Market Timing Optimization Model SYSTEMATIC

Calculates a composite Entry Score (0–1) from Momentum, Volatility, and Drawdown signals to determine optimal market entry zones. Full backtest vs Buy & Hold.

10 prompts · Momentum · Drawdown · Backtesting · CAGR Comparison



ML Direction Probability Model MACHINE LEARNING

Runs Logistic Regression, Random Forest, and Gradient Boosting simultaneously to predict next-day market direction. Interactive 3D probability surfaces with Bloomberg-style UI.

12 prompts · 3 ML Models · 3D Charts · Walk-Forward Accuracy



NIFTY 50 Macro Prediction Model MACRO / AI

Predicts NIFTY 50 monthly closing price using 43 global macro indicators (Dow Jones, Gold, Oil,

INR/USD, VIX, etc.). Uses Gradient Boosting, Random Forest, and Bayesian models trained on 18 years of data.

8 prompts · 43 Features · GB · RF · Student-t · 218 data points



Mean Reversion Signal Model STATISTICAL

Detects when prices deviate significantly from their statistical mean using Z-Score and Bollinger Bands. Generates buy/sell signals when reversion is statistically likely.

6 prompts · Z-Score · Bollinger Bands · RSI Divergence · Entry Signals



Pairs Trading Cointegration Model MARKET NEUTRAL

Finds statistically cointegrated stock pairs (e.g., RELIANCE vs TCS). Trades the spread when it diverges beyond a threshold and mean-reverts. Fully market-neutral strategy.

6 prompts · ADF Test · Cointegration · Spread Z-Score · Hedge Ratio



Risk Factor Attribution Model RISK MANAGEMENT

Decomposes portfolio returns into systematic risk factors: Market Beta, Size, Value, Momentum (Fama-French style). Identifies what truly drives your portfolio's performance.

5 prompts · Fama-French · Alpha/Beta · R-squared · Rolling Attribution



Value at Risk (VaR) Dashboard RISK / COMPLIANCE

Calculates Historical VaR, Parametric VaR, CVaR (Expected Shortfall), and Monte Carlo VaR for any portfolio. Stress-tests against 2008 Crash, COVID crash, and custom scenarios.

5 prompts · VaR · CVaR · Stress Test · 99% Confidence · Regulatory Format



Portfolio Optimization Engine PORTFOLIO THEORY

Finds the optimal portfolio allocation on the Efficient Frontier using Modern Portfolio Theory. Maximizes Sharpe Ratio or minimizes volatility for a basket of stocks.

5 prompts · Efficient Frontier · Sharpe Ratio · Max Return · Min Variance

SECTION 03 · MONTE CARLO SIMULATION ENGINE

Monte Carlo uses thousands of random price simulations based on historical drift and volatility. It answers "What is the range of possible outcomes?" rather than predicting a single price.

PROMPT
01

[BEGINNER] Basic Monte Carlo — NIFTY 50

You are a Python quant analyst. Run a Monte Carlo simulation for NIFTY 50. Use these parameters: Starting price: [CURRENT PRICE], Annual volatility: [VOL%], Annual drift: [DRIFT%], Simulations: 10,000, Horizon: 252 trading days. Plot all paths with a median line and 5th/95th percentile confidence band. Use Plotly for interactive charts. Show final price distribution histogram with VaR at 95% confidence.

 Replace [CURRENT PRICE] with today's NIFTY close, e.g. 23,500

PROMPT
02

[BEGINNER] Simulation with Full Dashboard

Build a complete Monte Carlo simulation dashboard for [NIFTY 50 / S&P 500 / TICKER] using Python and Plotly. Include: (1) 10,000 GBM paths plotted with percentile bands, (2) Final price distribution histogram, (3) Probability summary table: P(above target), P(below target), Expected value, VaR 95%, (4) Confidence interval at 50%, 75%, 90%, 95%. Display using subplot layout. Dark theme.

 Works for any instrument — swap NIFTY 50 for BTC-USD, GOOGL, etc.

PROMPT
03

[INTERMEDIATE] Blended Volatility — VIX + Historical

Build a Monte Carlo simulation for NIFTY 50 using blended volatility: $\text{blend_weight} = 0.6 \times \text{India_VIX}/100 + 0.4 \times \text{historical_vol_90day}$. Download NIFTY 50 (^NSEI) and India VIX (^INDIAVIX) data from yfinance. Run 10,000 simulations over 126 trading days. Show the blended vs pure historical comparison in a side-by-side chart. Use Plotly with dark Bloomberg theme.

 This is the professional-grade approach used by institutional desks

PROMPT
04

[ADVANCED] 3D Monte Carlo Paths Surface

Create a 3D Monte Carlo visualization for [NIFTY 50]. Download live data using yfinance. Compute 1,000 GBM paths. Plot a 3D surface where X=time (days), Y=simulation number, Z=price. Also create a second 3D density heatmap showing the probability mass over time. Use Plotly go.Surface and go.Heatmap. Add interactive sliders for number of simulations and time horizon.

 This visualization is ideal for Instagram/social media content — visually stunning

PROMPT

[ADVANCED] Portfolio Monte Carlo with Correlation

05

Build a correlated Portfolio Monte Carlo in Python for a 3-asset portfolio: [RELIANCE.NS, TCS.NS, INFY.NS] with weights [0.4, 0.35, 0.25]. Use Cholesky decomposition to preserve the historical correlation structure between assets. Run 10,000 portfolio simulations over 252 days. Show: (1) Efficient Frontier overlay, (2) Portfolio VaR vs individual VaR, (3) Correlation heatmap, (4) Portfolio growth paths. Plotly, dark theme.

 *Cholesky decomposition is the industry-standard method for correlated MC*

SECTION 04 · BLACK-SCHOLES OPTIONS PRICING ENGINE

Build a live NIFTY options pricing engine with all Greeks, interactive option chains, strategy builder, and 3D payoff surfaces. Works with real NSE weekly F&O data.

PROMPT
06

[BEGINNER] Basic Black-Scholes Pricer

Build a Python Black-Scholes option pricer for NIFTY 50 weekly options. Inputs: Spot=[CURRENT NIFTY], Strike=[STRIKE], Time to expiry=[DAYS/365], Risk-free rate=0.0525 (RBI rate), Volatility=India VIX/100. Output: CE price, PE price, and all 5 Greeks: Delta, Gamma, Theta, Vega, Rho. Format as a clean printed table. Also show the Put-Call parity check.

 RBI rate as of March 2026 is 5.25% — update if changed

PROMPT
07

[INTERMEDIATE] Live NIFTY Options Chain

Build an interactive NIFTY 50 options chain using Python that: (1) Downloads live NIFTY spot price from yfinance (^NSEI) and India VIX (^INDIAVIX), (2) Generates a full option chain across 21 strikes at ₹50 intervals (ATM ±10 strikes), (3) Prices each CE and PE using Black-Scholes with live VIX as volatility, (4) Displays Delta, Gamma, Theta, Vega for each strike, (5) Highlights ATM row in gold. Use Plotly interactive table. Dark theme.

 This generates a real-time options chain — update the spot price to get current data

PROMPT
08

[ADVANCED] 3D Volatility Surface

Generate a 3D Implied Volatility Surface for NIFTY 50 options in Python using Plotly. X-axis: Strike prices (21 strikes, ATM ±10 at ₹50 intervals). Y-axis: Time to expiry (7 days, 14 days, 21 days, 30 days, 45 days, 60 days). Z-axis: Implied volatility (simulate a realistic vol smile: higher IV for deep OTM/ITM, lower for ATM). Color by IV level. Rotate and zoom enabled. Dark Bloomberg theme with gold color scale.

 The vol smile/skew surface is the most important chart in options trading

PROMPT
09

[ADVANCED] Options Strategy Builder — P&L Payoff

Build an options strategy payoff diagram generator in Python for NIFTY 50. Include these 8 strategies: (1) Long Call, (2) Long Put, (3) Bull Call Spread, (4) Bear Put Spread, (5) Long Straddle, (6) Long Strangle, (7) Iron Condor, (8) Covered Call. For each: plot the P&L payoff diagram at expiry, show max profit / max loss / breakeven points, and calculate premium paid. Use Plotly with interactive strategy selector dropdown. NIFTY spot = [CURRENT PRICE], Lot size = 75.

 Iron Condor is the most popular neutral strategy for NIFTY weekly expiry traders

SECTION 05 · MARKET TIMING OPTIMIZATION MODEL

Determines WHEN to invest using a composite Entry Score from Momentum, Volatility, and Drawdown signals. Backtests the strategy against Buy & Hold.

PROMPT
10

[BEGINNER] Entry Score Calculator — Full Model

You are a Python quant analyst. Build a Market Timing model for NIFTY 50. Download 8 years of daily data from yfinance (^NSEI). Calculate three signals: (1) Momentum Signal: 20-day vs 200-day MA ratio, normalized 0-1, (2) Volatility Signal: inverse of 30-day realized vol, normalized 0-1, (3) Drawdown Signal: inverse of current drawdown from 52-week high, normalized 0-1. Combine: $\text{Entry_Score} = 0.4 \times \text{Momentum} + 0.35 \times \text{Volatility} + 0.25 \times \text{Drawdown}$. Plot the Entry Score over time with buy zones highlighted in green (score > 0.65). Dark theme, Plotly.

💡 The Entry Score tells you: above 0.65 = aggressive invest, 0.4–0.65 = moderate, below 0.4 = hold cash

PROMPT
11

[INTERMEDIATE] Backtest: Timing Strategy vs Buy & Hold

Extend the Market Timing model to include a full backtest. Strategy rules: Invest 100% when $\text{Entry_Score} > 0.60$, invest 50% when $0.40 < \text{Entry_Score} \leq 0.60$, hold cash (0%) when $\text{Entry_Score} \leq 0.40$. Starting capital: ₹10,00,000. Compare: Strategy Portfolio vs NIFTY Buy & Hold. Output: Total Return, CAGR, Max Drawdown, Sharpe Ratio, Number of Trades, Win Rate. Plot portfolio growth as dual line chart with drawdown ribbon. Plotly, dark Bloomberg theme.

💡 Real alpha comes from avoiding the big drawdowns — this model quantifies that edge

PROMPT
12

[ADVANCED] Rolling Regime Detection

Build a Market Regime Detection model for NIFTY 50 using Hidden Markov Models (HMM) with 3 regimes: Bull (low vol, positive drift), Bear (high vol, negative drift), and Sideways (mean-reverting). Use hmmlearn library. Download 10 years of NIFTY data. Overlay detected regimes on the price chart as colored backgrounds. Show regime probability timeline. Calculate: average return per regime, volatility per regime, and optimal position sizing per regime. Plotly interactive charts.

💡 HMM regime detection is used by hedge funds for dynamic position sizing

SECTION 06 · ML MARKET DIRECTION PROBABILITY

Predicts tomorrow's market direction (UP or DOWN) with probability scores using 3 machine learning models running simultaneously. Includes 3D interactive probability surfaces.

PROMPT

13

[INTERMEDIATE] Direction Probability Dashboard — 3 ML Models

Build a Market Direction Probability model for NIFTY 50 in Python. Download 5 years of daily data from yfinance. Features: 10-day return, 30-day return, RSI(14), 20-day volatility, 50-day MA distance, 200-day MA distance, India VIX level. Target: binary (1 = next day UP, 0 = next day DOWN). Train 3 models: Logistic Regression, Random Forest (200 trees), Gradient Boosting (XGBoost). Walk-forward validation with 252-day rolling window. Show: probability for next day (all 3 models), model accuracy comparison, feature importance bar chart, confusion matrices. Dark Bloomberg theme with ticker tape.

 Ensemble the 3 models by averaging their probabilities for a more robust signal

PROMPT

14

[ADVANCED] 3D Probability Surface — Feature Space

Create a 3D interactive Probability Surface for NIFTY 50 direction prediction. Train a Gradient Boosting model on 5 years of data. Plot a 3D surface where X-axis = RSI(14), Y-axis = 30-day Momentum, Z-axis = P(UP) from the model. Color the surface by probability (red = high P(DOWN), green = high P(UP)). Add a scatter cloud of actual data points colored by actual outcome. Rotation and zoom enabled. Plotly, dark theme. Also show model accuracy, AUC-ROC score.

 This 3D surface is the viral chart style — shows the full decision boundary of the model

SECTION 07 · NIFTY 50 MACRO PREDICTION MODEL

Predicts monthly NIFTY 50 closing price using 43 global macro indicators. Trained on 18 years of data covering multiple market cycles including the 2008 crash and COVID.

PROMPT

15

[ADVANCED] Macro Model — Full Build

I have an Excel file with 218 rows (Jan 2008 to Feb 2026) and 44 columns: MONTH (date), NIFTY (target), and 43 global macro features: Dow Jones, S&P 500, NASDAQ, FTSE, Nikkei, Hang Seng, Gold, Silver, WTI Oil, Brent Oil, Natural Gas, Copper, INR/USD, EUR/USD, GBP/USD, USD/JPY, India 10Y yield, US 10Y yield, Fed Funds Rate, India VIX, CBOE VIX, India CPI, US CPI, India IIP, FII Net Flows, DII Net Flows, India PMI, US PMI, China PMI, India M3 Money Supply, and more. Build a lag-1 prediction model: given month T-1 values of all 43 indicators, predict NIFTY close at month T. Train: Gradient Boosting, Random Forest, and Bayesian Ridge. Output: next month NIFTY prediction, confidence interval, feature importance chart, actual vs predicted backtest chart. Python, Plotly, dark theme.

 [Upload your Excel data file to Google Colab first, then run this prompt](#)

PROMPT

16

[INTERMEDIATE] Macro Correlation Heatmap

Build a macro correlation heatmap for NIFTY 50 using Python. Create a synthetic dataset (or use your Excel file) with NIFTY and 43 global macro indicators. Calculate the Pearson correlation of each indicator with NIFTY close (lagged by 1 month). Plot as an interactive heatmap sorted by correlation strength. Highlight the top 10 most correlated and top 10 most inversely correlated features. Use Plotly with a diverging red-green color scale. Show p-values to indicate statistical significance.

 [This shows which global factors actually drive NIFTY — essential research](#)

SECTION 08 · NEW MODELS (Exclusive to This Guide)

These 5 models are brand-new additions created exclusively for this master guide by Deepkumar Khinchi. They extend the existing library with advanced risk management and alpha generation strategies.

NEW MODEL 1: Mean Reversion Signal Engine

Mean reversion is the statistical tendency of prices to return toward their average after extreme moves. This model detects overextended price levels and generates high-probability entry signals.

PROMPT 17 [NEW] Mean Reversion Z-Score Model

Build a Mean Reversion Signal model for NIFTY 50 in Python. Download 5 years of data from yfinance. Calculate: (1) Rolling Z-Score = $(\text{price} - \text{rolling_mean_20}) / \text{rolling_std_20}$, (2) Bollinger Band position = $(\text{price} - \text{lower_band}) / (\text{upper_band} - \text{lower_band})$, (3) RSI(14) divergence from 50. Generate BUY signals when: Z-Score < -1.5 AND RSI < 40 AND price near lower Bollinger Band. Generate SELL/EXIT signals when Z-Score > 0 OR RSI > 60. Backtest with ₹10L capital. Plot: price with signals marked, Z-Score over time, equity curve. Show: win rate, average holding period, Sharpe ratio. Plotly dark theme.

 Mean reversion is especially powerful in index instruments like NIFTY during sideways markets

NEW MODEL 2: Pairs Trading Cointegration Model

Pairs trading is a market-neutral strategy that profits from the temporary divergence of two historically correlated stocks. No market direction bet — pure statistical arbitrage.

PROMPT 18 [NEW] Pairs Trading — NIFTY Stocks

Build a Pairs Trading model for Indian stocks using Python. Step 1: Test for cointegration between these pairs using the Engle-Granger test: (RELIANCE.NS, ONGC.NS), (TCS.NS, INFY.NS), (HDFCBANK.NS, ICICIBANK.NS). Step 2: For the most cointegrated pair, calculate the optimal hedge ratio using OLS regression. Step 3: Calculate the spread Z-Score = $(\text{spread} - \text{mean}) / \text{std}$ over 60-day rolling window. Step 4: Trade rules: LONG spread when $Z < -2$, SHORT spread when $Z > +2$, EXIT when $|Z| < 0.5$. Backtest with ₹20L capital, 50/50 split. Output: spread Z-Score chart with entry/exit, P&L chart, Sharpe ratio, max drawdown, number of trades. Plotly, dark theme, download 5 years of data.

 This strategy was used by Long-Term Capital Management (LTCM) — learn the mechanics

NEW MODEL 3: Risk Factor Attribution (Fama-French Style)

Understand what actually drives your portfolio returns by decomposing them into systematic risk factors. This is how institutional risk managers evaluate funds and portfolios.

PROMPT
19

[NEW] Factor Attribution — Portfolio Decomposition

Build a Risk Factor Attribution model in Python for a sample portfolio: [RELIANCE.NS 30%, TCS.NS 25%, HDFCBANK.NS 20%, INFY.NS 15%, HDFC.NS 10%]. Download 3 years of daily data from yfinance. Define 4 factors: (1) Market Factor: NIFTY 50 daily returns (^NSEI), (2) Momentum Factor: 12-month return minus 1-month return, (3) Volatility Factor: inverse of 30-day realized vol, (4) Quality Factor: synthetic quality score based on return consistency. Run rolling 90-day OLS regression of portfolio returns on the 4 factors. Plot: (1) Rolling factor loadings (beta to each factor), (2) Factor contribution to portfolio return, (3) Residual (alpha) after factor explanation, (4) R-squared over time. Plotly interactive, dark theme.

 This tells you exactly WHERE your returns are coming from — factor beta or true alpha

NEW MODEL 4: VaR & Stress Testing Dashboard

Value at Risk (VaR) quantifies the maximum expected loss over a given time period at a specified confidence level. Required by regulators for institutional risk management.

PROMPT
20

[NEW] Multi-Method VaR Dashboard

Build a comprehensive Value at Risk dashboard in Python for a NIFTY 50 portfolio. Download 10 years of NIFTY data. Calculate 4 types of VaR at 95% and 99% confidence for a 1-day horizon: (1) Historical VaR: percentile of historical return distribution, (2) Parametric VaR: $-z \times \sigma \times P$ (assumes normal distribution), (3) CVaR/Expected Shortfall: average loss beyond VaR threshold, (4) Monte Carlo VaR: 50,000 GBM simulations. Add Stress Testing scenarios: (a) 2008 Crash: -60% shock, (b) COVID March 2020: -38% shock, (c) Custom shock: user-defined %. Display: side-by-side comparison table of all VaR methods, return distribution histogram with VaR marked, rolling 252-day VaR chart, stress test bar chart. ₹1 Crore portfolio. Plotly, dark theme.

 Regulatory requirement: SEBI requires institutional managers to report VaR daily

NEW MODEL 5: Efficient Frontier Portfolio Optimizer

Find the mathematically optimal allocation of capital across a basket of stocks that maximizes return per unit of risk — the foundation of Modern Portfolio Theory.

PROMPT
21

[NEW] Efficient Frontier — Portfolio Optimization

Build a Portfolio Optimization model in Python using Modern Portfolio Theory. Stocks: [RELIANCE.NS, TCS.NS, HDFCBANK.NS, INFY.NS, ITC.NS, TATAMOTORS.NS]. Download 3 years of daily returns from yfinance. Step 1: Generate 50,000 random portfolio weight combinations. Step 2: For each portfolio, calculate: Expected Annual Return, Annual Volatility

(std), Sharpe Ratio (risk-free rate = 5.25%). Step 3: Plot the Efficient Frontier — scatter chart of all portfolios colored by Sharpe Ratio (Plotly). Step 4: Mark: Maximum Sharpe Ratio portfolio (star), Minimum Variance portfolio (diamond), Equal Weight portfolio (circle). Step 5: Display optimal weights as a pie chart. Step 6: Show the Capital Market Line. Dark theme, interactive hover showing weights on each dot.

🔗 *The max Sharpe portfolio is mathematically the best risk-adjusted allocation — this finds it automatically*

SECTION 09 · QUICK REFERENCE & TIPS

Common Python Libraries Used

 Library	 What It Does
yfinance	Download market data (NSE/BSE/NYSE)
pandas	Data manipulation and analysis
numpy	Numerical computations and GBM
plotly	Interactive 3D charts and dashboards
scipy	Statistical tests (ADF, cointegration)
scikit-learn	ML models (RF, LR, GB)
xgboost	Gradient Boosting models
hmmlearn	Hidden Markov Models for regimes

Key Financial Formulas Reference

 Formula	 Description
GBM: $S(t) = S_0 \times \exp((\mu - \sigma^2/2)t + \sigma\sqrt{t} \times Z)$	Monte Carlo simulation formula
$VaR_{95} = -1.645 \times \sigma \times \text{Portfolio_Value}$	Parametric VaR at 95% confidence
$\text{Sharpe} = (R_p - R_f) / \sigma_p$	Risk-adjusted return metric
Black-Scholes CE = $S \times N(d1) - K \times e^{(-rT)} \times N(d2)$	European call option pricing
Z-Score = $(X - \mu) / \sigma$	Standard deviations from mean
$\text{CAGR} = (\text{End}/\text{Start})^{(1/\text{Years})} - 1$	Compound Annual Growth Rate

NSE India Key Parameters (March 2026)

 Parameter	 Value
RBI Repo Rate	5.25% per annum
NIFTY 50 Lot Size	75 units per contract
Weekly Expiry Day	Thursday (weekly)
NIFTY Ticker (yfinance)	^NSEI

India VIX Ticker

^INDIAVIX

NSE Trading Hours

9:15 AM – 3:30 PM IST

SECTION 10 · ABOUT THE AUTHOR

ABOUT THE AUTHOR

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Quantitative Finance Educator & Content Creator

Deepkumar Khinchi is the creator of @deepthinksfinance — a platform dedicated to making quantitative finance accessible to everyone through AI-powered tools, educational content, and ready-to-use model guides. With a passion for blending mathematics, machine learning, and market analysis, Deepkumar has built a growing community of quant finance enthusiasts, traders, and students across India and globally.

This Master Prompt Guide represents the culmination of months of research, model-building, and community feedback — all shared freely for the financial education of every aspiring quant.

 Instagram · @deepthinksfinance

Follow for more quant models, prompt guides, and financial education content

"Making Quant Finance Accessible — One Model at a Time"



10 Models Built

Complete model library from basics to expert level



50+ Prompts

Copy-paste ready for Claude, ChatGPT & Gemini



100% Free

All tools work on Google Colab — no paid software needed

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MARKET TIMING OPTIMIZATION QUANT FINANCE MODEL

Quantitative Finance Guide

By @deepthinksfinance

STEP-BY-STEP PROMPT GUIDE

For Complete Beginners — Google Colab Edition

Build your own AI-powered model that tells you WHEN to invest in the market

WHAT YOU WILL BUILD

A complete Quantitative Finance model using Python that:

- Downloads 8 years of real NIFTY 50 market data automatically
- Calculates Momentum, Volatility & Drawdown signals
- Creates a composite Entry Score to time the market
- Runs a backtest comparing your strategy vs Buy & Hold
- Shows interactive charts of portfolio growth & buy zones
- Outputs Total Return, CAGR & Max Drawdown metrics

SECTION 1 — INTRODUCTION

1.1 What Is Market Timing?

Market timing is the practice of deciding WHEN to enter or exit the stock market based on data, rather than staying fully invested at all times. Instead of asking 'Which stock should I buy?', this model answers:

"When is the best time to put my money into the market?"

1.2 What Is This Model? (Plain English)

This model reads NIFTY 50 price data every day and calculates three signals:

Signal	What It Measures
Momentum	Is the market trending upward or downward?
Volatility	How wild / risky is the market right now?
Drawdown	How far has the market fallen from its peak?

It then combines these into a single Entry Score (0 to 1). A high score means 'good time to invest heavily.' A low score means 'be cautious.'

1.3 Why Use Google Colab?

Google Colab is a free, browser-based Python notebook. You do not need to install anything on your computer. All you need is a Google account.

- Free to use — no software installation required
- Runs in your browser — works on any device
- All Python libraries are pre-installed or easily added
- You can share your notebook with anyone via a link

1.4 Prerequisites — What You Need Before Starting

You do NOT need prior coding experience. This guide will walk you through every step. However, make sure you have:

1. A Google Account (Gmail) — free to create at google.com
2. Access to Google Colab — visit colab.research.google.com
3. A stable internet connection (downloads market data live)
4. About 30 minutes of uninterrupted time

SECTION 2 — SETTING UP GOOGLE COLAB

2.1 Opening Google Colab

Follow these steps exactly to open a new notebook:

5. Open your browser and go to: `colab.research.google.com`
6. Sign in with your Google account if prompted
7. Click 'New Notebook' (blue button, top left)
8. A new untitled notebook will open — you will see an empty code cell

2.2 Understanding the Colab Interface

Colab is made up of 'cells'. Each cell can contain code or text. Here is what you need to know:

Part of Interface	What It Does
Code Cell	Where you type/paste Python code
Play Button (▶)	Runs the code in that cell
+ Code	Adds a new code cell below
Runtime menu	Controls the Python environment
Output area	Shows results, charts, and errors below each cell

TIP: To run a cell, click the Play (▶) button on its left side, or press Shift + Enter on your keyboard.

2.3 Installing Required Libraries

Before running the main model, you need to install one library that is not pre-installed in Colab. Create your FIRST code cell and paste this exactly:

```
# CELL 1 — Install required library
# Paste this into your first Colab cell and press the Play (▶) button

!pip install yfinance --quiet
```

After you run this cell, you should see a success message. The other libraries (numpy, pandas, matplotlib, plotly) are already available in Colab by default.

IMPORTANT: You must run Cell 1 first, every time you open a new Colab session. The libraries reset when you close the tab.

SECTION 3 — THE MASTER PROMPT (COPY & PASTE)

3.1 What Is a Prompt?

In this context, a 'prompt' is a complete set of instructions you give to an AI (like Gemini, ChatGPT, or Claude) to generate Python code for you. You do NOT write the code yourself — the AI writes it based on the prompt.

Alternatively, you can paste the prompt directly into Google Colab as a comment and let Colab AI (powered by Gemini) generate the code for you inline.

3.2 Two Ways to Use This Prompt

Method	How It Works
Method A — AI Chatbot	Copy the prompt → paste into ChatGPT / Gemini / Claude → copy the generated code → paste into Colab
Method B — Colab AI	Open Colab → click the AI button (sparkle icon) → paste the prompt → let Colab write code for you directly

3.3 The Complete Master Prompt

Copy EVERYTHING below (from the first line to the last line) and paste it into your AI tool of choice:

```
Act as a Quantitative Finance Researcher.
Build a Market Timing Optimization Model for NIFTY 50 using 8 years of daily data.

Use Python with: yfinance, numpy, pandas, matplotlib, plotly

OBJECTIVE:
Create a model that tells WHEN to invest, not WHAT to buy.

STEP 1 — DATA:
Download 8 years of daily data for ^NSEI (NIFTY 50) using yfinance.

STEP 2 — FEATURES:
```


Calculate:

- 50-day moving average (MA50)
- 200-day moving average (MA200)
- Momentum Strength = $(\text{Price} / \text{MA200}) - 1$
- 20-day rolling volatility (std dev of daily returns), normalized
- Rolling maximum price
- Drawdown = $(\text{Price} - \text{Rolling Max}) / \text{Rolling Max}$

STEP 3 – ENTRY SCORE:

Create a composite Entry Score by combining:

- Momentum component (positive momentum = higher score)
- Volatility component (lower volatility = higher score)
- Drawdown component (deeper drawdown = buying opportunity)

Normalize all 3 components to 0-1 range. Combine them equally.

STEP 4 – POSITION SIZING STRATEGY:

Based on Entry Score (no binary buy/sell signals):

- High Score (above 0.6) --> 100% invested
- Medium Score (0.3-0.6) --> 50% invested
- Low Score (below 0.3) --> 15% invested

STEP 5 – BACKTEST:

Simulate INR 100,000 starting capital.

Calculate daily portfolio value based on position sizing.

Compare Strategy vs Buy & Hold over the same period.

STEP 6 – VISUALS (use plotly for all charts):

Plot 1: Portfolio growth – Strategy vs Buy & Hold

Plot 2: Entry Score over time with threshold lines

Plot 3: Drawdown chart for both strategies

Plot 4: NIFTY price with Buy Zones highlighted (green = high score)

STEP 7 – OUTPUT:

Print a performance summary showing:

- Total Return % (Strategy vs Buy & Hold)
- CAGR % (Strategy vs Buy & Hold)
- Max Drawdown % (Strategy vs Buy & Hold)
- Sharpe Ratio (Strategy vs Buy & Hold)

BONUS (include this): Add ipywidgets sliders for:

- Momentum weight (0.0 to 1.0)
- Volatility weight (0.0 to 1.0)
- Drawdown weight (0.0 to 1.0)

Weights should auto-normalize. Show updated results interactively.

Output clean, well-commented Python code ready to run in Google Colab.

Use clear section headers as comments in the code.

SECTION 4 — STEP-BY-STEP EXPLANATION FOR BEGINNERS

This section explains each part of the model in plain English so you understand what is happening at every stage.

STEP 1

Downloading NIFTY 50 Data

What Happens

The model uses yfinance (Yahoo Finance library) to automatically download 8 years of NIFTY 50 closing prices. The ticker symbol for NIFTY 50 on Yahoo Finance is ^NSEI.

Why 8 Years?

8 years of data gives the model enough history to capture multiple market cycles — bull runs, corrections, crashes, and recoveries — making the strategy more reliable.

Beginner Prompt Tip

If the data download fails, try asking the AI: *"Fix the yfinance download error for ^NSEI and add error handling."*

STEP 2

Calculating the Three Key Features

Feature 1 — Momentum

Momentum measures the strength of the market trend. We calculate two Moving Averages (MA):

Indicator	Meaning
MA50 (50-day)	Average price of last 50 days — short-term trend
MA200 (200-day)	Average price of last 200 days — long-term trend
Momentum Strength	$(\text{Current Price} / \text{MA200}) - 1$ = how far above/below trend

If Momentum Strength is positive (price is above MA200), the market is in an uptrend. If negative, the market is below its long-term average.

Feature 2 — Volatility

Volatility measures how much the market is swinging up and down. We calculate the standard deviation of daily returns over the last 20 days. High volatility = risky, uncertain market. Low volatility = stable, calmer market.

Note: We INVERT the volatility signal — lower volatility gets a HIGHER score, because it represents a less risky entry point.

Feature 3 — Drawdown

Drawdown measures how far the market has fallen from its recent peak. Formula:

$$\text{Drawdown} = (\text{Current Price} - \text{Rolling Maximum Price}) / \text{Rolling Maximum Price}$$

A deep drawdown (e.g., -30%) means the market has crashed significantly. Historically, deep drawdowns represent buying opportunities for patient investors.

STEP 3 Calculating the Entry Score

The Entry Score combines all three features into a single number between 0 and 1. Here is how it works:

Score Range	Market Interpretation
0.7 — 1.0	Excellent time to invest (strong uptrend, low volatility, recovering from drawdown)
0.4 — 0.7	Moderate opportunity (mixed signals, partial investment)
0.0 — 0.4	Caution advised (high risk, high volatility, or weak trend)

Each component is normalized to a 0-1 scale and then averaged (or weighted if you use the sliders). This prevents any single feature from dominating the score.

STEP 4 Position Sizing Strategy

Instead of a binary Buy / Sell signal, the model uses position sizing — it adjusts HOW MUCH of your capital is invested based on the Entry Score:

Entry Score	Action Taken
Above 0.6 (High)	100% of capital invested — maximum confidence
0.3 to 0.6 (Medium)	50% of capital invested — cautious participation
Below 0.3 (Low)	15% of capital only — minimal risk exposure

KEY INSIGHT: You are always in the market to some degree (15% minimum). This prevents missing sudden sharp recoveries while still protecting you from major downturns.

STEP 5 Running the Backtest

The backtest simulates what would have happened if you used this model over the past 8 years with a starting capital of INR 1,00,000. It calculates:

- Daily portfolio value based on the Entry Score-driven position sizing
- A Buy & Hold portfolio (always 100% invested, no timing) for comparison
- The difference in final value between the two strategies

IMPORTANT DISCLAIMER: Past performance does not guarantee future results. This model is for educational purposes only and is not financial advice.

STEP 6 The Visualizations

The model generates 4 interactive charts using Plotly. You can zoom, pan, and hover over them in Colab:

Chart	What It Shows
Portfolio Growth	Strategy vs Buy & Hold value over time (INR)
Entry Score	The composite score over time with threshold lines at 0.3 and 0.6
Drawdown Chart	How much each strategy fell from its peak at any point
NIFTY Buy Zones	Price chart with green highlights showing when score was high

STEP 7 Performance Output

After the charts, the model prints a clean summary table. Here are the four metrics and what they mean:

Metric	Plain English Explanation
Total Return %	Total profit as a % of starting capital over the full period
CAGR %	Compounded Annual Growth Rate — your average yearly return
Max Drawdown %	Worst peak-to-trough loss — how much you lost before recovering
Sharpe Ratio	Return per unit of risk — higher is better (above 1.0 is good)

SECTION 5 — BONUS: INTERACTIVE SLIDERS

5.1 What Are the Sliders?

The bonus section adds ipywidgets sliders — interactive controls you can drag with your mouse directly inside the Colab notebook. They let you change the weight of each signal and instantly see how results change.

5.2 The Three Slider Weights

Slider	What It Controls
Momentum Weight	How much importance to give to trend direction
Volatility Weight	How much importance to give to market stability
Drawdown Weight	How much importance to give to buying-the-dip opportunities

The three weights are automatically normalized, so they always sum to 1.0. For example, if you set all three to 1.0, each gets 33.3% weight.

5.3 Experimenting With Weights — Suggested Presets

Try these configurations to understand how each signal affects performance:

Preset Name	Momentum / Volatility / Drawdown
Balanced (default)	1.0 / 1.0 / 1.0 — equal weight to all three
Trend Follower	2.0 / 0.5 / 0.5 — focus on uptrends only
Contrarian / Dip Buyer	0.5 / 0.5 / 2.0 — buy when market has crashed
Risk Averse	1.0 / 2.0 / 1.0 — avoid entering during volatile periods

SECTION 6 — TROUBLESHOOTING COMMON ERRORS

6.1 Common Issues and Fixes

If you run into problems, here are the most common errors and exactly what to type to the AI to fix them:

Error / Problem	What To Ask the AI
ModuleNotFoundError: yfinance	"Fix: install yfinance in Google Colab and re-run the download"
Download returns empty data	"The yfinance download for ^NSEI returned empty. Fix it with a date range from 2016 to today"
Chart does not display	"The plotly chart is not showing in Colab. Use fig.show() and add renderer='colab'"
Sliders not working	"The ipywidgets sliders are not interactive in Colab. Fix the widget display"
NaN errors in score	"I am getting NaN values in the Entry Score. Add .dropna() and forward-fill missing values"
Code runs but no output	"Add print statements after each section to confirm data is loading correctly"

6.2 How to Re-Run After Errors

9. Go to Runtime menu → Restart and Run All
10. If that fails, go to Runtime → Factory Reset Runtime
11. Then run Cell 1 (pip install) again before running the main code

SECTION 7 — UNDERSTANDING THE MODEL DEEPER

7.1 When Does the Model Invest?

The model invests heavily when ALL THREE of the following are true simultaneously:

- The market is above its 200-day moving average (uptrend / momentum is strong)
- Recent daily price swings are small (volatility is low — stable market)
- The market has already corrected from a recent peak (drawdown creates opportunity)

When all three conditions align, the Entry Score rises above 0.6 and the model puts 100% to work.

7.2 When Does the Model Avoid the Market?

- Market is below its 200-day MA (bearish trend — momentum is weak or negative)
- Very high day-to-day price swings (panic / uncertainty in the market)
- Market is at an all-time high with no recent correction (no margin of safety)

7.3 Why Do Volatility and Drawdown Matter?

Together, volatility and drawdown act as a safety filter:

High volatility often precedes or accompanies market crashes. By reducing exposure when volatility is high, the model avoids getting caught in sharp sell-offs.

Deep drawdowns, on the other hand, historically mark the best long-term buying opportunities. The 2020 COVID crash, the 2016 correction, and the 2022 bear market were all followed by strong recoveries. The model identifies these as high-score zones.

7.4 Limitations of This Model

- This model uses hindsight (historical data) — future results may differ significantly

- Transaction costs (brokerage, STT, slippage) are not included in the backtest
- The model does not account for dividends or corporate actions
- It works on index-level data only — not individual stocks
- No model is perfect — combine this with your own research and risk tolerance

SECTION 8 — NEXT STEPS & ADVANCED IDEAS

8.1 Modifications to Try

Once you have the model running, try these prompt additions to make it more advanced:

- Add RSI (Relative Strength Index) as a 4th feature to the Entry Score
- Add MACD signal line crosses as an additional confirmation filter
- Test the model on SENSEX (^BSESN) and compare results to NIFTY
- Add a rebalancing rule — only change position monthly, not daily
- Include transaction cost simulation (0.1% per trade both ways)

8.2 Prompts to Extend the Model

Here are ready-to-use follow-up prompts to improve your model:

Add RSI:

```
Add a 14-day RSI indicator to the existing market timing model.  
Include it as a 4th component of the Entry Score.  
RSI below 40 should increase the score (oversold = buying opportunity).  
RSI above 70 should decrease the score (overbought = reduce exposure).  
Normalize RSI to 0-1 and add a slider for its weight.
```

Add Transaction Costs:

```
Modify the backtest to include transaction costs.  
Charge 0.1% of portfolio value whenever position size changes by more than 10%.  
Show how total return changes with and without transaction costs.
```

8.3 Sharing Your Notebook

12. In Colab, go to File → Share
13. Set access to 'Anyone with the link can view'
14. Copy the link and share it with others

QUICK REFERENCE CARD

3-STEP QUICK START

Step	Action
1. Open Colab	Go to colab.research.google.com → New Notebook
2. Install Library	Paste: <code>!pip install yfinance --quiet</code> → Run
3. Run the Prompt	Copy Section 3.3 prompt → paste into AI → run generated code

GLOSSARY OF KEY TERMS

Term	Plain English Definition
yfinance	Python library to download stock price data from Yahoo Finance
Moving Average (MA)	Average price over last N days — smooths out noise
Momentum	Speed and direction of market price movement
Volatility	How much prices are swinging — higher = riskier
Drawdown	How far the market has fallen from its recent highest point
Entry Score	Combined 0-1 signal telling you how good the current entry point is
Position Sizing	Deciding what % of your capital to invest (not all-or-nothing)
Backtest	Testing a strategy on past data to see how it would have performed
CAGR	Compound Annual Growth Rate — average yearly % return
Sharpe Ratio	Measures return vs risk — higher is better
ipywidgets	Python library for interactive sliders in Jupyter / Colab notebooks

DISCLAIMER

This guide is for educational and informational purposes only. Nothing in this document constitutes financial advice. Always consult a SEBI-registered financial advisor before making investment decisions. Past model performance does not guarantee future results.

THE COMPLETE

MONTE CARLO SIMULATION

PROMPT GUIDE FOR CLAUDE AI

20 Ready-to-Copy Prompts · GBM / VaR / Options · 3D Dashboards · Live Market Data
NIFTY 50 · S&P 500 · Stocks · Crypto · Portfolios

20 Prompts

6 Levels

Copy & Use

Mar 2026

Built with Claude AI · Designed for quant finance enthusiasts, traders & content creators.

Paste any prompt directly into Claude. Swap the [BRACKETS] for your own instrument, timeframe, or data.

CONTENTS

What's Inside

Navigate to any section quickly

SECTION	WHAT YOU WILL BUILD
Section 1 — Foundations	The GBM formula explained · Drift & Volatility · What Monte Carlo actually does
Section 2 — Starter Prompts	5 beginner prompts: basic simulation, histogram, confidence bands, probability table
Section 3 — Intermediate	Multi-variable blended vol · Historical vs implied VIX · Regime-aware simulation
Section 4 — Advanced 3D	3D Monte Carlo paths · Density surface · Scatter cloud · Heatmap (Plotly)
Section 5 — Live Market	Prompts that search the web for live price, VIX, FII/DII, FX rate before simulating
Section 6 — Variations	Portfolio MC · Options pricing · Crypto · Intraday · Multi-asset correlated paths
Section 7 — Customisation	How to modify any prompt · Variable cheat-sheet · Style & theme guide
Section 8 — Formula Ref.	GBM · Log Returns · VaR · Sharpe · Black-Scholes · All the maths in one place

SECTION 1

Monte Carlo Foundations

What it is, how it works, why it matters for markets

What Is a Monte Carlo Simulation?

A Monte Carlo simulation runs thousands of possible future price paths for a financial asset, each driven by random chance within historical statistical bounds. The result isn't a single prediction — it's a probability distribution of outcomes.

Think of it like this: instead of asking 'where will NIFTY be in 30 days?', you ask 'across 10,000 possible futures, what does the range of NIFTY prices look like?' That range tells you far more than any single forecast.

The Core Formula — Geometric Brownian Motion (GBM)

Σ **GBM DAILY PRICE STEP**

$$\text{Price}(t+1) = \text{Price}(t) \times \exp((\mu - 0.5\sigma^2) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z)$$

SYMBOL	MEANING
μ (mu)	Mean daily return (drift) — estimated from historical log returns
σ (sigma)	Daily volatility — std dev of historical log returns
Δt	Time step = 1 (one trading day)
Z	Random draw from Standard Normal distribution $N(0,1)$
$\exp(...)$	Exponential function — ensures price can never go below zero
$P(t+1)$	Simulated price tomorrow given today's price $P(t)$

The Three Core Parameters

📌 **NOTE**

Before writing any Monte Carlo prompt, you need three inputs: a starting price, a drift (μ), and a volatility (σ). Everything else flows from these.

PARAMETER	WHERE TO GET IT
Starting Price	Latest closing price from your data or a live web search
Daily Drift (μ)	Mean of log returns = $\text{mean}(\ln(P_t / P_{t-1}))$ from historical data
Daily Volatility (σ)	Std dev of log returns from historical data — OR use $VIX/100/\sqrt{252}$ for implied vol

SECTION 2

Starter Prompts

5 beginner-friendly prompts — copy, paste, get results

PROMPT 01 [BEGINNER] Minimal Monte Carlo Simulation

You are a Python quant analyst. Run a Monte Carlo simulation for [NIFTY 50 / S&P 500 / TICKER].

Parameters:

- Current price: [24,158]
- Annual volatility: [22%]
- Annual drift: [9%]
- Simulations: 10,000
- Horizon: 30 trading days

Use the GBM formula:

$$\text{Price}(t+1) = \text{Price}(t) * \exp((\mu - 0.5 * \sigma^2) * dt + \sigma * \sqrt{dt} * Z)$$

Convert annual to daily: $\mu_{\text{daily}} = \mu / 252$, $\sigma_{\text{daily}} = \sigma / \sqrt{252}$

Print: Expected price, Median, P5, P95, P(gain), P(loss)

Then generate a clean HTML page showing a paths chart and results table.

 **CUSTOMISE:** Replace [24,158] with today's price. Replace [22%] / [9%] with your instrument's real volatility and expected annual return.

PROMPT 02 [BEGINNER] Simulation with Histogram & Distribution

You are a quantitative finance analyst. Run a Monte Carlo simulation for [INSTRUMENT]:

- Starting price: [PRICE]
- Historical annual volatility: [VOL%]
- Annual drift: [DRIFT%]
- Simulations: 10,000 paths, Horizon: 30 days

After simulating, produce an HTML page with TWO Plotly charts:

Chart 1 — PATHS: Plot 200 sample paths as thin translucent lines.

Color paths green if final price > start, red if below.

Draw the mean path as a thick white line.

Chart 2 — HISTOGRAM: Distribution of final prices at day 30.

Color bars: red if price < current-5%, blue if within 5%, green if > +5%

Add vertical dashed lines for: Current, Expected, P5, P95

Label each line.

Style: Dark background (#050A0F), monospace font, no axis borders.

 **CUSTOMISE:** The two-chart combo is the most shareable format. Great for social media posts.

PROMPT 03 [BEGINNER] Confidence Bands Chart

Build a confidence interval band chart for a Monte Carlo simulation of [INSTRUMENT].

Run 10,000 GBM paths. Starting price [PRICE], daily mu = [MU], daily sigma = [SIGMA]. Horizon: 30 trading days.

Compute at each time step: P5, P25, Mean, P75, P95 paths.

Generate HTML with a single Plotly chart:

- Shaded area between P5 and P95 (colour: rgba(0,180,255,0.08))
- Shaded area between P25 and P75 (colour: rgba(0,180,255,0.15))
- Mean path: bright cyan, width 3
- P5 line: dashed red, P95 line: dashed green
- Horizontal dashed white line at current price
- X-axis: 'Trading Day', Y-axis: 'Price (₹)'
- Dark theme: paper_bgcolor='#050A0F', plot_bgcolor='#050A0F'
- Add annotation showing expected price at day 30

 **CUSTOMISE:** This is the clearest chart for showing 'where will price be'. Use it for explainer content.

PROMPT 04 [BEGINNER] Probability Summary Table

Run a Monte Carlo simulation for [INSTRUMENT] with:

- Start price: [PRICE], Annual vol: [VOL%], Annual drift: [DRIFT%]
- 10,000 simulations, 30 trading days

After simulating, calculate ALL of these probabilities:

- P(Price increases from start)
- P(Price decreases from start)
- P(+2% gain) P(-2% loss)
- P(+5% gain) P(-5% loss)
- P(+10% gain) P(-10% loss)
- P(above [KEY LEVEL 1]) e.g. above 25,000
- P(below [KEY LEVEL 2]) e.g. below 23,000

Display as an HTML table with:

- Left column: Scenario name
- Middle column: Target price
- Right column: Probability % (green for bullish, red for bearish)

Dark theme, bold font for probabilities, hover highlighting on rows.

 **CUSTOMISE:** Replace [KEY LEVEL 1] and [KEY LEVEL 2] with psychologically important price levels: round numbers, 52-week highs/lows, support/resistance.

PROMPT 05 [BEGINNER] Full Beginner Dashboard (All-in-One)

You are a quant analyst and Python developer. Build a complete Monte Carlo dashboard for [NIFTY 50].

Data: Historical monthly prices (provide your data or describe columns here)

Steps:

1. Load data → sort by date → compute log returns
2. Estimate: mu (mean daily return), sigma (daily std dev)

3. Run 10,000 GBM simulations over 30 trading days
4. Calculate: Expected price, Median, P5/P25/P75/P95, all probabilities

Build a single HTML file with FOUR sections:

SECTION A — Market Summary: Current price, Annual vol %, Annual drift %

SECTION B — Simulation Results: A table of all percentile prices + changes from current

SECTION C — Probability Grid: Colour-coded cards showing key probabilities

SECTION D — Charts: Paths chart + Histogram + Confidence bands (all Plotly, dark theme)

Include a short 3-sentence interpretation at the bottom.

Make it look like a Bloomberg terminal. Output complete HTML only.

 **CUSTOMISE:** This is the prompt that built the NIFTY dashboard you saw. Start here, then add 3D charts using Section 4 prompts.

SECTION 3

Intermediate Prompts

*Blended volatility, VIX regimes, realistic parameter estimation***PROMPT 06 [INTERMEDIATE] Blended Volatility (VIX + Historical)**

Build a Monte Carlo simulation for [NIFTY 50] using blended volatility:

Inputs:

- Current price: [24,158]
- India VIX (implied annual vol): [23.36%]
- Historical annual vol from 18yr data: [20.14%]
- Blend ratio: 70% implied + 30% historical
- Blended sigma_{annual} = $0.70 * (VIX/100) + 0.30 * \text{hist_vol}$

Convert to daily:

- $\mu_{\text{daily}} = \text{annual_drift} / 252$
- $\sigma_{\text{daily}} = \text{blended_sigma_annual} / \sqrt{252}$

Run 10,000 GBM simulations over [15] trading days.

In the dashboard, show a side-by-side comparison:

- Historical vol simulation results vs Blended vol results
- Which gives wider confidence interval and why

Explain: 'Current VIX of 23.36% signals elevated fear – the market is pricing in more risk than the 18-year average.'

 **CUSTOMISE:** Always use blended vol when VIX is elevated (>20). Historical vol alone understates current market risk during stressed periods.

PROMPT 07 [INTERMEDIATE] Volatility Regime-Aware Simulation

Build a regime-aware Monte Carlo simulation for [INSTRUMENT].

Step 1 – Identify volatility regimes from historical data:

- LOW regime: Annual vol < 15% (calm markets)
- MEDIUM regime: Annual vol 15-25% (normal markets)
- HIGH regime: Annual vol > 25% (stressed markets)
- CRISIS regime: Annual vol > 35% (crash / GFC-like)

Step 2 – Estimate separate mu and sigma for each regime from historical data.

Step 3 – Current regime: [HIGH – VIX at 23.36]

Run simulation using HIGH-regime parameters.

Also run with MEDIUM-regime parameters as a 'recovery scenario'.

Step 4 – Dashboard shows:

- Regime statistics table (4 regimes × avg return, vol, P(gain) in each)
- Two side-by-side fan charts: current regime vs recovery scenario
- Regime history chart: shaded bands showing which regime was active each year
- Interpretation: 'In HIGH volatility regimes historically, NIFTY has...'

 **CUSTOMISE:** This is the most realistic simulation approach. It stops you from using calm-period parameters during a crisis.

PROMPT 08 [INTERMEDIATE] Parameter Sensitivity Analysis

Build a Monte Carlo sensitivity dashboard for [INSTRUMENT].

Base case: Start = [24,158], mu = [0.04%/day], sigma = [1.41%/day]

Run the simulation 9 times varying parameters:

Sigma scenarios: Low vol [0.8%], Base [1.41%], High vol [2.2%]

Mu scenarios: Bear [-0.05%/day], Neutral [0.04%], Bull [0.12%/day]

Create a 3x3 grid of mini-charts (Plotly subplots) showing the confidence bands for each parameter combination. Title each: e.g. "Low Vol + Bear Drift"

Also build a sensitivity table:

Rows = Sigma scenarios, Columns = Mu scenarios

Each cell shows: Expected Price | P(Gain) | P5 | P95

Highlight the BASE CASE cell in cyan. Color cells green if P(gain)>55%, red if <45%.

 **CUSTOMISE:** This prompt is great for content — the 3x3 grid shows your audience how vol and drift interact. Very visual, very shareable.

PROMPT 09 [INTERMEDIATE] Rolling Monte Carlo (Backtest Accuracy)

You are a quant analyst. Backtest the accuracy of Monte Carlo simulations using historical data for [NIFTY 50].

Method — Rolling window backtest:

1. For each month from Jan 2010 to Oct 2024:

- Use the previous 36 months of data to estimate mu and sigma
- Run a Monte Carlo 30-day simulation from that month's start price
- Record: Predicted P5, Mean, P95 at month-end
- Record: Actual price at month-end

2. Calculate coverage statistics:

- How often did actual price fall within P5-P95? (should be ~90%)
- Average error: $|\text{Predicted mean} - \text{Actual}| / \text{Actual} * 100$
- Bias: Was simulation systematically too high or too low?

3. Dashboard shows:

- Time series of predicted range vs actual price
- Coverage ratio chart (rolling 12-month)
- Error distribution histogram
- Summary: 'The GBM model captured [X]% of outcomes within the 90% CI'

 **CUSTOMISE:** This is the most rigorous prompt in the guide. It proves — or disproves — how accurate Monte Carlo is on your specific instrument.

SECTION 4

Advanced 3D Interactive Charts

3D Monte Carlo visualisations using Plotly — drag, rotate, zoom

NOTE

All 3D charts below use Plotly's scatter3d and surface trace types. They are fully interactive in a browser — drag to rotate, scroll to zoom, hover for values.

PROMPT 10 [ADVANCED] 3D Monte Carlo Path Tornado

Build a 3D Monte Carlo path chart for [INSTRUMENT] — the 'tornado' view.

Run 10,000 GBM simulations. Sample 500 paths for 3D rendering.

Create a Plotly scatter3d chart:

X-axis = Trading Day (0 to 30)
Y-axis = Simulation index (0 to 499)
Z-axis = Simulated Price

Path colouring:

- Final price > current: colour = rgba(0,255,100, 0.06) — faint green
- Final price < current: colour = rgba(255,50,50, 0.06) — faint red
- Path opacity: 0.08, line width: 0.8

Special overlays:

- Mean path: cyan, width = 6, plotted at Y = 250 (midpoint)
- P95 path: dashed green, width = 3, Y = 490
- P5 path: dashed red, width = 3, Y = 10

Scene: dark background #030610, cyan grid lines, camera eye at x=1.8 y=1.2 z=0.9

Title: 'NIFTY 50 — 10,000 GBM Simulations · 30 Trading Days'

Add subtitle annotation: 'Drag to rotate · Scroll to zoom'

 **CUSTOMISE:** The tornado chart is the most impressive visualisation for social media. The coloured paths create a stunning fan effect.

PROMPT 11 [ADVANCED] 3D Probability Density Surface

Build a 3D probability density surface — showing HOW LIKELY each price is at each future date.

Run 10,000 GBM simulations for [INSTRUMENT].

At each time step $t = 0, 1, 2, \dots, 30$:

Compute a histogram (40 bins) of simulated prices at that step.
Normalise to probability density (density=True in numpy).

Build a Plotly surface3d chart:

X-axis = Trading Day (0 to 30)

Y-axis = Price bins (P1 to P99 of final prices, 40 bins)
Z-axis = Probability density

Colourscale (dark to bright as density increases):
[0, '#030610'], [0.2, '#003060'], [0.5, '#006090'],
[0.75, '#00C0D0'], [0.9, '#00FF88'], [1.0, '#FFFF00']

Enable contour projection on the Z floor.
Add a horizontal plane at Z=0 (transparent base).
Camera: eye at x=1.8, y=-1.5, z=1.2 for best wave view.
The chart should look like an ocean wave — narrow at day 0, spreading wide by day 30.

 **CUSTOMISE:** This is the 'wow' chart. The expanding wave shows uncertainty growing over time. Perfect for explaining why forecast ranges widen.

PROMPT 12 [ADVANCED] 3D Scatter Cloud (Outcome Space)

Create a 3D scatter cloud visualising the full Monte Carlo outcome space for [INSTRUMENT].

Run 10,000 simulations. Sample 500 paths × every 3rd day = ~2,500 data points.

Each point represents (Simulation #, Day, Price at that day).

Plotly scatter3d with marker:
X = Simulation index, Y = Day, Z = Price
Size = 1.5 (small dots for density effect)
Colour = Price value (Z), using diverging colourscale:
Low prices → red (#FF2D55)
Mid prices → gold (#FFCC00)
High prices → cyan (#00F5FF)
Very high → green (#00FF88)
Opacity = 0.65

The cloud should be dense in the middle (near current price) and thin at extremes.
Add 3 vertical reference planes at P5, Mean, P95 prices using mesh3d.
Dark background. Camera at x=1.6 y=1.6 z=1.0.

 **CUSTOMISE:** Combine this with the density surface in a tabbed dashboard for maximum impact. The contrast between the two 3D charts is stunning.

PROMPT 13 [ADVANCED] 2D Heatmap (Price-Time Density)

Build a 2D heatmap of simulation density for [INSTRUMENT].

Run 10,000 GBM simulations over 30 days.

At each time step, bin prices into 50 buckets between P1 and P99.
Create a 2D density matrix: rows = price bins, cols = time steps.
Normalise each column so the sum = 1 (conditional density at each day).

Plotly heatmap:
X = Day (0 to 30), Y = Price bins

Z = Density values
Colourscale: dark blue (low) → cyan → green → yellow (high density)
No gap between cells.

Overlay three lines on top of the heatmap:

- White dashed: Current price (horizontal)
- Cyan solid: Mean path
- Red dashed: P5 path, Green dashed: P95 path

Add annotation at day 30 showing expected price.

Use dark background. This chart looks like a thermal imaging scan.

 **CUSTOMISE:** The heatmap is the most 'data-dense' chart — every pixel carries information. Great for advanced/technical audiences.

SECTION 5

Live Market Data Prompts

Prompts that search the web for real-time data before simulating

⚠ WARNING

These prompts ask Claude to search the web for live data. Always verify any fetched prices and statistics before using them for real decisions.

PROMPT 14 [LIVE DATA] Live NIFTY Simulation (Web Search)

You are a quant analyst. Before running the simulation, SEARCH THE WEB for current data on NIFTY 50:

Search for:

1. "NIFTY 50 current price today" → get current price
2. "India VIX today" → get implied volatility
3. "FII DII flows March 2026" → get institutional flow data
4. "USD INR rate today" → get Rupee exchange rate
5. "NIFTY 52 week high low 2026" → get range context

Once you have live data, run a Monte Carlo simulation:

- Starting price: [fetched live price]
- Blended vol: $70\% * (\text{India VIX}/100) + 30\% * 0.2014$ (historical)
- Daily drift: 0.000356 (from 18-year historical average)
- 10,000 simulations, 15 trading days (rest of current month)

Build HTML dashboard with:

1. Live data ticker tape (all fetched values)
2. Simulation results with current market context callout box
3. Monte Carlo paths + distribution charts (Plotly, dark theme)
4. FII vs DII context: 'FII selling [amount] but DII buying [amount]...'

🔧 CUSTOMISE: Add 'Search as of [today's date]' in your prompt to help Claude find the most recent data.

PROMPT 15 [LIVE DATA] Any Global Index — Auto-Fetch & Simulate

Search the web for current data on [S&P 500 / NASDAQ / DAX / FTSE / NIKKEI]:

Fetch:

- Current index level and today's % change
- Current CBOE VIX (or equivalent fear gauge)
- Year-to-date return
- 52-week range

Then build a Monte Carlo simulation:

- Start: fetched current level
- Sigma: $(\text{fetched VIX} / 100) / \sqrt{252}$ for daily vol
- Mu: based on 20-year historical average annual return for this index
(S&P 500: ~10%, NASDAQ: ~12%, DAX: ~9%, FTSE: ~7%, NIKKEI: ~8%)

- 10,000 simulations, 21 trading days (1 month)

Dashboard must show:

- Flag/icon for the index country
- All fetched live statistics
- Simulation fan chart with confidence bands
- Key probabilities: $P(\text{index above [round number] by month-end})$
- Brief macro commentary based on current VIX level

 **CUSTOMISE:** Works for any major index. Change the historical return assumption to match the index's long-run average.

SECTION 6

Simulation Variations

*Portfolio, options, crypto, intraday — extend the core model***PROMPT 16 [PORTFOLIO] Correlated Multi-Asset Portfolio**

Build a correlated Monte Carlo portfolio simulation for a portfolio of:

- NIFTY 50: [50%] weight
- Gold: [20%] weight
- US Bonds: [15%] weight
- USD/INR FX: [15%] weight

Using historical monthly data provided:

1. Calculate the 4x4 correlation matrix between log returns
2. Use Cholesky decomposition to generate correlated random shocks:

$$Z_{\text{correlated}} = \text{Cholesky}(\text{Correlation_Matrix}) \times Z_{\text{independent}}$$
3. Simulate each asset's daily price path using GBM with correlated Z
4. Compute daily portfolio value = sum of weighted asset values

Run 10,000 portfolio paths over 30 days. Calculate:

- Portfolio VaR (95%) and CVaR
- Portfolio Expected Return and Std Dev
- Probability of portfolio loss > [5%]
- Diversification benefit: compare portfolio vol vs weighted-average vol

Dashboard: correlation heatmap, portfolio paths fan chart, risk metrics table.

 **CUSTOMISE:** Cholesky decomposition ensures your simulated assets move together realistically. Without it, assets are independent — unrealistic.

PROMPT 17 [OPTIONS] Monte Carlo for Options Pricing

Use Monte Carlo simulation to price a European [CALL/PUT] option on [NIFTY 50]:

Option details:

- Current underlying price S: [24,158]
- Strike price K: [25,000]
- Time to expiry T: [21 trading days / 252]
- Risk-free rate r: [6.5% / 252] per day (India 10Y bond)
- Implied volatility sigma: [23.36% / sqrt(252)] per day

Simulation steps (risk-neutral pricing):

Use drift = r (risk-free rate), NOT historical drift
 Run 100,000 paths (options pricing needs more paths for accuracy)
 At expiry compute payoff:
 CALL: $\max(S_T - K, 0)$
 PUT: $\max(K - S_T, 0)$
 Discount: $\text{Option_Price} = \text{mean}(\text{payoffs}) \times \exp(-r \times T)$

Compare with Black-Scholes analytical price.

Show: MC price, BS price, difference, convergence chart (price vs N simulations),

Greeks estimated numerically from MC. HTML output, dark theme.

 **CUSTOMISE:** Always use risk-neutral drift (r) not historical drift for options pricing. Using historical drift gives wrong prices.

PROMPT 18 [CRYPTO] Crypto Monte Carlo (Fat Tails)

Build a Monte Carlo simulation for [BITCOIN / ETH / any crypto] accounting for fat tails.

Problem with standard GBM for crypto: Crypto has extreme moves (>5 sigma) far more often than Normal distribution predicts.

Solution — use Student-t distribution instead of Normal:

```
Z_fat = t.rvs(df=3) / sqrt(3/(3-2))    # normalise so std dev = 1
(degrees of freedom = 3 gives heavy tails matching crypto behaviour)
```

Inputs: Current BTC price $[\$[PRICE]]$, daily vol sigma = [3.5%], daily drift = [0.05%]

Run 10,000 simulations for [30] days using fat-tail Z.

Also run with Normal Z. Compare results side-by-side:

- Fat-tail P5 will be MUCH lower than Normal P5
- Fat-tail P95 will be MUCH higher than Normal P95

Dashboard shows both simulations, with commentary:

'Using Student-t (df=3): P5 = $\$X$ vs Normal: P5 = $\$Y$ — a [N]% difference in tail risk'

 **CUSTOMISE:** For stocks use Normal. For crypto, small-caps, or during crises — always use fat-tail distributions. The difference is enormous at the tails.

PROMPT 19 [INTRADAY] Intraday Monte Carlo (5-min data)

Build an intraday Monte Carlo simulation using [NIFTY / STOCK] 5-minute data.

Data: 5-minute OHLCV data for the last [20 trading days]

Steps:

1. Calculate 5-minute log returns
2. Estimate: μ_{5min} (mean 5-min return), σ_{5min} (std dev of 5-min returns)
3. Note: intraday has U-shaped volatility (higher at open and close)
 - Segment the day into: Open (9:15-10:15), Midday (10:15-14:30), Close (14:30-15:30)
 - Estimate separate sigma for each segment.

Run 5,000 intraday GBM simulations for rest of today ([N] remaining 5-min bars).

Use the appropriate segment sigma at each time step.

Show:

- Intraday path fan chart (X = time, not day number)
- Probability of hitting [key intraday level] before close
- Expected closing price with confidence range
- Compare: flat-vol GBM vs segmented-vol GBM

 **CUSTOMISE:** Intraday MC requires much more data (5-min bars). Without U-shape vol adjustment, the simulation is unrealistic at market open and close.

PROMPT 20 [LONG-TERM] Monthly Return Distribution for Year-End Forecast

Build a year-end Monte Carlo forecast for [NIFTY 50] using monthly returns.

Why monthly? For 12-month forecasts, monthly returns are less noisy than daily.

From 18-year historical monthly data:

1. Compute monthly log returns
2. Estimate: μ_{monthly} , σ_{monthly}
3. Convert: Annual drift = $\mu_{\text{monthly}} * 12$, Annual vol = $\sigma_{\text{monthly}} * \sqrt{12}$

Run 10,000 simulations of 12 monthly steps (not 252 daily steps).

Additional analysis:

- What are the chances of ending December [2026] above [28,000]?
- What are the chances of a 20%+ correction at any point during the year?
(For this, track the monthly minimum across each path, not just the final value)
- Show the distribution of calendar-year returns (histogram of annual % change)
- Compare: How does current starting level compare to historical valuation?

Dashboard: yearly fan chart, annual return distribution, key level probabilities.

 **CUSTOMISE:** For year-end forecasts, track intermediate highs and lows — not just the final price. An investor cares if the portfolio drops 20% during the year, even if it recovers.

SECTION 7

Customisation Guide

How to adapt every prompt to your instrument and style

The [BRACKET] Variables — Full Cheat Sheet

BRACKET	WHAT TO REPLACE IT WITH
[INSTRUMENT]	NIFTY 50 · Bank Nifty · Reliance · TCS · S&P 500 · Bitcoin · EUR/USD
[PRICE]	Latest closing price. Get from NSE website, TradingView, or ask Claude to search.
[VOL%]	Annual historical volatility. Rule of thumb: Nifty ~20%, S&P 500 ~18%, BTC ~70%
[DRIFT%]	Expected annual return. Use 8-10% for large-cap indices, adjust for growth stocks.
[VIX VALUE]	India VIX for Indian markets. CBOE VIX for US. Use today's value for live accuracy.
[SIMULATIONS]	1,000 = fast preview. 10,000 = standard. 50,000 = high precision options pricing.
[DAYS]	30 = 1 month. 21 = 1 month of trading days. 63 = 1 quarter. 252 = 1 year.
[KEY LEVEL]	Round numbers: 25,000. 52-week high/low. Previous ATH. Strong support/resistance.
[THEME]	Dark Bloomberg / Cyberpunk Dark / Clean White / Print-Friendly / Neon Terminal
[CHART TYPE]	Plotly (interactive) / Matplotlib (static PNG) / Chart.js (web embed)

Style & Theme Guide

THEME	PROMPT PHRASE TO ADD
Dark Bloomberg	Use dark background #050A0F, cyan accents #00D4FF, monospace fonts (DM Mono), minimal borders
Cyberpunk Neon	Use black background, neon green and pink glows, scanline overlay, Orbitron or Share Tech Mono font
Clean White	Use white background, navy text, grey borders, professional sans-serif font, no glow effects
Print-Friendly	White background, all text black or dark grey, no Plotly (use Matplotlib static PNG instead)
Social Media Card	Fixed 1200×630px container, large numbers, bold typography, strong contrast, screenshot-ready

Weak vs Strong Prompt Examples

× WEAK	✓ STRONG
"Simulate the stock price"	"Run 10,000 GBM simulations for NIFTY 50 starting at ₹24,158 with daily sigma = 1.41% and daily mu = 0.036%, over 30 trading days"
"Make a nice chart"	"Plotly scatter3d: 500 paths, X=day, Y=sim index, Z=price. Colour green if final > start else red. Mean path in cyan width=5. Dark bg #030610"
"Add volatility"	"Use blended vol: sigma = 0.70*(India_VIX/100/sqrt(252)) + 0.30*(0.2014/sqrt(252)). Show both implied-only and blended results side by side"



PRO TIP

The more specific your numbers, the better the output. Always provide: exact starting price, exact sigma and mu values, exact number of simulations, exact horizon in trading days, and exact chart specification including colours.

SECTION 8

Formula Reference

Every equation you need — copy into any prompt

Core Monte Carlo Formulas

Σ GBM — Daily Price Step

$$P(t+1) = P(t) \times \exp((\mu - 0.5\sigma^2) \cdot 1 + \sigma \cdot Z) \quad \text{where } Z \sim N(0,1)$$

Σ Log Return (always use this, not simple return)

$$r_t = \ln(P_t / P_{t-1}) \rightarrow \text{daily log return}$$

Σ Parameter Estimation from Historical Data

$$\mu_{\text{daily}} = \text{mean}(r_t) \quad | \quad \sigma_{\text{daily}} = \text{std}(r_t)$$

Σ Annualisation

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252} \quad | \quad \mu_{\text{annual}} = \mu_{\text{daily}} \times 252$$

Σ Blended Volatility (VIX + Historical)

$$\sigma_{\text{blend}} = 0.70 \times (\text{VIX}/100 \div \sqrt{252}) + 0.30 \times \sigma_{\text{historical_daily}}$$

Σ Value at Risk (95%)

$$\text{VaR}_{95} = -\text{percentile}(\text{final_prices}, 5\%) \quad [\text{expressed as loss from start}]$$

Σ Conditional VaR / Expected Shortfall

$$\text{CVaR}_{95} = -\text{mean}(\text{final_prices}[\text{final_prices} < \text{VaR}_{95}])$$

Σ Sharpe Ratio

$$\text{Sharpe} = (\text{Portfolio_Return} - \text{Risk_Free_Rate}) \div \text{Portfolio_StdDev}$$

Σ Black-Scholes Call Price (for comparison vs MC)

$$\text{Call} = S \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2) \quad d1 = [\ln(S/K) + (r + 0.5\sigma^2)T] \div (\sigma\sqrt{T}) \quad d2 = d1 - \sigma\sqrt{T}$$

Σ Cholesky Decomposition (Correlated Returns)

$L = \text{cholesky}(\Sigma) \rightarrow Z_{\text{corr}} = L \times Z_{\text{indep}}$ where Σ is the correlation matrix between assets

Monte Carlo Parameter Quick Reference

INSTRUMENT	TYPICAL PARAMETERS (approximate starting values)
NIFTY 50	Annual vol ~20%, Annual drift ~9-10%, VIX typically 12-18 (elevated: >20)
Bank Nifty	Annual vol ~25-28%, Higher beta to NIFTY, more volatile — use $VIX \times 1.3$
S&P 500	Annual vol ~16-18%, Annual drift ~10%, VIX typically 15-20 (elevated: >25)
NASDAQ 100	Annual vol ~22-25%, Annual drift ~12-14%, Higher tech premium
Gold (XAUUSD)	Annual vol ~12-15%, Annual drift ~6-8%, Negative correlation to equities
Crude Oil (WTI)	Annual vol ~30-40%, High drift uncertainty — use shorter horizons
Bitcoin (BTC)	Annual vol ~60-80%, No reliable drift — use short horizons only
EUR/USD	Annual vol ~7-9%, Near-zero drift for FX pairs, use interest rate diff

You now have 20 prompts to build anything.

Start with Prompt 01. Build the basics. Then layer in 3D charts, live data, and portfolio features.

Share your dashboards, tag the source, and help your followers discover quantitative finance.

⚠ DISCLAIMER: Educational use only. Not financial advice. All investments carry risk.

Portfolio Optimization Quant Model Master Prompt

A complete, copy-paste template for building a professional portfolio optimization system using AI. Replace the securities list with your favorite stocks and let AI generate the entire quantitative model.

How to Use This Guide

For Independent Traders

Paste this prompt into your preferred AI model, replace the stock symbols with your portfolio universe, and get a complete portfolio optimization system built automatically.

No coding expertise required - the AI generates production-ready Python code.

The Complete Master Prompt

```
You are a quantitative finance researcher and Python developer. Create a clean, well-structured Python script that builds a portfolio optimization model using Monte Carlo simulation. The model should analyze a portfolio of stocks using the last 3 years of market data.

Use the following Python libraries:
yfinance
numpy
pandas
scipy
matplotlib
plotly

SECURITIES
Use the following stocks as the portfolio universe:
RELIANCE.NS
TCS.NS
HDFCBANK.NS
ICICIBANK.NS
INFY.NS
ITC.NS
LT.NS
AXISBANK.NS
SBIN.NS
BHARTIARTL.NS
KOTAKBANK.NS

MODEL OBJECTIVE
Build a quantitative portfolio optimization model that finds the optimal portfolio allocation based on risk and return.

STEP 1 — DATA COLLECTION
Download 3 years of historical daily adjusted close prices for the selected securities using yfinance.
Clean the data and remove missing values.

STEP 2 — RETURN CALCULATION
Calculate:
- Daily returns
- Annualized returns
- Annualized volatility
Assume 252 trading days in a year.

STEP 3 — RISK STRUCTURE
Calculate:
- Covariance matrix
- Correlation matrix

STEP 4 — MONTE CARLO PORTFOLIO SIMULATION
Generate at least 5000 random portfolios.
For each portfolio calculate:
- Expected annual return
- Portfolio volatility
- Sharpe ratio
Store the results in a dataframe.

STEP 5 — OPTIMAL PORTFOLIOS
Identify:
- Maximum Sharpe Ratio Portfolio
- Minimum Volatility Portfolio
Display their portfolio weights.

STEP 6 — EFFICIENT FRONTIER
Plot the efficient frontier using a scatter plot:
- X axis = portfolio volatility
- Y axis = expected return
- Color the points by Sharpe ratio.
- Highlight the optimal portfolios.

STEP 7 — PORTFOLIO BACKTEST
Simulate how an investment of ₹100000 would have grown over the last 3 years using:
- Maximum Sharpe portfolio
- Minimum volatility portfolio
- Equal weight portfolio
Plot portfolio growth over time.

OUTPUT
Display:
- Optimal portfolio weights
- Expected return
- Portfolio volatility
- Sharpe ratio

Also show the following charts:
- Efficient Frontier
- Portfolio Allocation Pie Chart
- Portfolio Growth Simulation
```



Customizing the Securities List

Replace the example securities with any stocks or ETFs you want to analyze. The model works with any combination of tickers from major exchanges worldwide.

US Stocks Example

AAPL MSFT NVDA TSLA GOOG AMZN META NFLX

ETF Portfolio Example

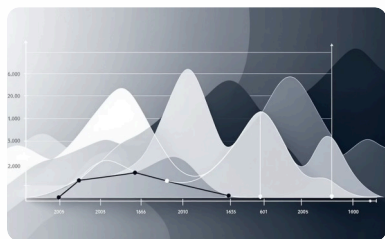
SPY QQQ EEM AGG BND VTI

International Mix

AAPL TSLA BABA RELIANCE.NS TCS.NS

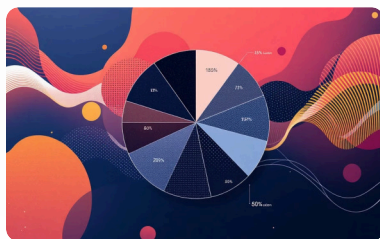
- ❏ **Pro Tip:** Include 5-15 securities for best results. Too few securities limits diversification analysis. Too many securities can make optimization less effective and visualizations harder to interpret.

What the Model Creates



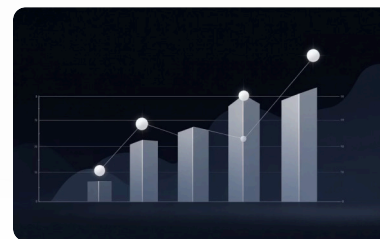
Efficient Frontier

Visual representation of optimal risk-return trade-offs showing 5,000+ simulated portfolios



Portfolio Allocation

Pie charts displaying optimal weight distribution across selected securities



Growth Simulation

Backtested performance comparing optimal portfolios against equal weighting

The model generates comprehensive visualizations and quantitative analysis that professional portfolio managers use daily. Each chart serves a specific purpose in evaluating portfolio efficiency and risk characteristics.

Expected Output Components

01

Optimal Portfolio Weights

Percentage allocation for each security in both maximum Sharpe and minimum volatility portfolios

02

Performance Metrics

Expected annual return, portfolio volatility, and Sharpe ratio for each optimal portfolio configuration

03

Risk Analysis

Covariance and correlation matrices showing relationships between securities in your portfolio

04

Visual Charts

Interactive plots including efficient frontier scatter plot, allocation pie charts, and growth simulation

The output combines quantitative metrics with visual analysis, providing both numerical precision and intuitive understanding of portfolio characteristics. This dual approach mirrors professional quant research methodology.

Technical Implementation Details

Monte Carlo Simulation

Generates 5,000+ random portfolio weight combinations to map the risk-return landscape. Each simulation calculates expected return, volatility, and Sharpe ratio.

- Random weight generation with sum-to-one constraint
- Annualized return calculation using 252 trading days
- Volatility computed from covariance matrix
- Sharpe ratio assumes risk-free rate of 0%

Data Pipeline

Automatically downloads and processes 3 years of historical price data using yfinance library.

- Daily adjusted closing prices only
- Missing data removal with forward-fill
- Return calculation with percentage change
- Covariance matrix for risk modeling



Common Issues & Troubleshooting

1

Invalid Ticker Symbols

Verify all securities use correct exchange suffixes (e.g., .NS for NSE, .L for LSE). Check spelling and capitalization. Some tickers may have changed or been delisted.

2

Rate Limit Exceeded

yfinance has rate limits. If you get errors, reduce the number of securities or add delay logic. Consider caching data to disk between runs.

3

Missing Data Points

Different securities have different listing dates. The model automatically aligns data to common date range. Results may be biased if some stocks have limited history.

4

Convergence Issues

Monte Carlo requires sufficient simulations. If optimal portfolios seem unstable, increase simulation count from 5,000 to 10,000 or 20,000.



Pro Tip: For production use, add error handling around data download and implement data validation checks before running optimization. Include logging to track progress through each step.

Next Steps & Enhancements



Add Constraints

Include minimum/maximum weight limits, sector constraints, or ESG filters



Dynamic Rebalancing

Extend to multi-period optimization with rebalancing rules and transaction costs



Risk Management

Incorporate Value-at-Risk, conditional statistics, or downside risk measures

This master prompt provides a solid foundation for portfolio optimization. For advanced users, consider extending the model with additional risk metrics, incorporating transaction costs, adding position sizing rules, or integrating with live trading systems. The modular structure makes it easy to build upon this base implementation.

Q U A N T I T A T I V E F I N A N C E · N S E I N D I A

Black-Scholes Pricing Engine

COMPLETE PROMPT GUIDE FOR CLAUDE AI

How to ask Claude to build, extend & interpret your NIFTY options model

BLACK-SCHOLES
PRICING ENGINE

NSE · NIFTY 50
WEEKLY F&O

LOT SIZE: 75
UNITS/CONTRACT

RBI RATE: 5.25%
MARCH 2026

01

SECTION

Introduction

What this tool is and how Claude AI makes it interactive

What is the NIFTY Live Options Lab?

The NIFTY Live Options Lab is a fully interactive, browser-based quantitative finance platform that implements the Black-Scholes option pricing model specifically for NSE NIFTY 50 index options. It fetches live market data from Yahoo Finance, computes real-time option prices, displays 3D interactive surfaces, and provides a complete strategy builder with payoff diagrams.

Claude AI can help you build every component of this model, extend it, debug it, interpret its outputs, and customise it for professional use. This guide provides ready-to-use prompts organised by the five main areas of the tool.

What the Tool Contains

- Live NIFTY 50 spot price from Yahoo Finance (^NSEI)
- India VIX (^INDIAVIX) — the NSE fear gauge
- Black-Scholes CE/PE pricing engine with all five Greeks
- Interactive option chain across 21 strikes (₹50 intervals)
- Strategy builder: 8 strategies with 3D payoff surfaces
- Five 3D Plotly surfaces: Spot×Vol, Spot×Time, Vol Smile, Greeks, Delta
- Analysis tab: Theta decay, Delta profile, Probability cone

HOW TO USE THIS GUIDE

Copy any prompt exactly as shown in the blue shaded cells and paste it into Claude AI. Claude will generate Python code, JavaScript, or explanations depending on what you ask. Always start with context-setting prompts before complex requests.

02

SECTION

Black-Scholes Formula Prompts

Ask Claude to implement, explain and extend the BS model

2.1 Core Formula Implementation

Use these prompts to have Claude implement the Black-Scholes formula in Python (Google Colab) or JavaScript (browser-based). The tool uses a JavaScript implementation for real-time slider interactivity.

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Implement BS in Python	Act as a quant finance researcher. Implement the Black-Scholes option pricing formula in Python for NIFTY 50 index options. Use $S=23000$, $K=23000$, $T=7/365$, $r=0.0525$, $\sigma=0.165$. Show Call price, Put price, $d1$, $d2$, and all five Greeks using <code>scipy.stats.norm</code> .	Full Python function with all outputs, formatted like a Colab notebook
JS real-time version	Convert the Black-Scholes formula into a JavaScript function that runs in the browser. It should accept S , K , T_days , r_pct , σ_pct and return call, put, delta, gamma, theta, vega, rho. Use an Abramowitz & Stegun approximation for <code>normCDF</code> .	Self-contained JS function ready to drop into HTML
Verify Put-Call Parity	Given NIFTY spot= 23000 , $K=23000$, $T=7$ days, $r=5.25\%$, $\sigma=16.5\%$, compute the Black-Scholes call and put prices and verify Put-Call Parity: $C - P = S - K \cdot e^{-rT}$. Show the numerical check.	Numerical verification with LHS vs RHS comparison
Explain $d1$ and $d2$ intuitively	Explain $d1$ and $d2$ in the Black-Scholes formula in plain English with a NIFTY 50 example. What do $N(d1)$ and $N(d2)$ represent economically? What happens to $d1$ as volatility increases?	Intuitive explanation with NIFTY-specific numbers
ATM option pricing	For NIFTY ATM options ($S=K=23000$) with 7 DTE, $r=5.25\%$, and India VIX= 16.5% , compute the theoretical straddle price (call + put). What is the expected daily move implied by this straddle?	Straddle price + implied daily move calculation

2.2 Input Parameters — NIFTY Context

These prompts help you correctly set and interpret the five Black-Scholes inputs for NIFTY index options specifically.

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Current NIFTY parameters	What realistic Black-Scholes input parameters should I use for NIFTY 50 options as of March 2026? Include: spot price range, typical implied volatility from India VIX, RBI risk-free rate, and weekly vs monthly expiry in years.	Parameter table with typical ranges and sources

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Annualise India VIX	India VIX is quoted as an annualised percentage. If India VIX = 16.5, what is the daily implied move of NIFTY? What is the weekly implied move? Show the formula used.	Daily and weekly move calculations with formulas
Convert DTE to years	NIFTY has weekly expiry every Thursday. How do I correctly convert days-to-expiry (DTE) into the T parameter for Black-Scholes? Should I use calendar days or trading days? Show examples for 7 DTE and 30 DTE.	Correct T conversion with calendar vs trading day comparison
Risk-free rate for India	What risk-free rate should I use in Black-Scholes for NIFTY options? The RBI repo rate is 5.25%. Should I use repo rate, 91-day T-bill yield, or OIS rate? What is the market convention on NSE?	Explanation of Indian risk-free rate conventions

03

SECTION

Option Chain Prompts

Build and interpret the NSE NIFTY option chain

3.1 Building the Option Chain

The option chain displays theoretical CE/PE prices across 21 strikes ($\text{ATM} \pm 10 \times ₹50$) for any selected expiry. Use these prompts to build or extend the chain logic.

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Generate full option chain	Build a NIFTY option chain in Python. For spot=23000, strikes from 21000 to 25000 in ₹50 steps, T=7 days, r=5.25%, sigma=16.5%, compute using Black-Scholes: CE price, PE price, CE delta, PE delta, gamma, theta, vega for each strike. Output as a pandas DataFrame.	pandas DataFrame with all chain columns
ATM strike selection	Given NIFTY spot at 23,147, what is the ATM strike? How do I programmatically find the ATM strike when strikes are in ₹50 intervals? Show Python code that rounds to the nearest ₹50 strike.	Python ATM selection logic with rounding
Add OI simulation	For my NIFTY option chain, add simulated open interest (OI) data. OI should peak at ATM and decrease further OTM/ITM. Calls have higher OI below spot (ITM calls / OTM puts) and puts have higher OI above spot. Show the Python function.	Realistic OI distribution function for NIFTY
Moneyness classification	For each strike in a NIFTY option chain, classify the option as: Deep ITM, ITM, Near ATM (within 0.5%), ATM, Near OTM, OTM, Deep OTM. What are the standard moneyness thresholds used on NSE?	Moneyness classification function with NSE thresholds
Expiry date logic	NIFTY has weekly expiry every Thursday and monthly expiry on the last Thursday. Write Python code that given today's date (March 19, 2026), computes the next 5 expiry dates and their DTE values.	Python expiry date calculator for NSE NIFTY

3.2 Reading & Interpreting the Chain

Once the chain is displayed, use these prompts to understand what the data means and how to make trading decisions from it.

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Interpret PCR	NIFTY put-call ratio (PCR) based on OI is 1.35. What does this mean? Is it bullish or bearish? What PCR range is considered neutral, bullish, and bearish for NIFTY?	PCR interpretation with NIFTY-specific thresholds

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Max pain calculation	Explain the concept of max pain for NIFTY options. Given the option chain DataFrame (CE prices, PE prices, all strikes), write Python code to compute the max pain strike – the price at which option sellers lose the least.	Max pain algorithm in Python
IV skew analysis	In the NIFTY option chain, OTM puts have higher implied volatility than OTM calls. This is the volatility skew. Explain why NIFTY puts are more expensive and how to quantify the skew using the 25-delta risk reversal.	Skew explanation with risk reversal calculation
Support/resistance from chain	How do option traders use NIFTY option chain data to identify support and resistance levels? What strike with the highest put OI is likely to act as support? Show an example with Python code.	Support/resistance identification from OI data

04

SECTION

Option Greeks Prompts

Understand and use all five Greeks for NIFTY risk management

4.1 Greeks Reference Table

GREEK	RANGE	MEASURES	NIFTY CONTEXT	FILL
Δ Delta	Call: 0–1 Put: -1–0	Price sensitivity to spot move	ITM=0.70 means ₹0.70 gain per ₹1 NIFTY rise	
Γ Gamma	Always positive	Rate of change of Delta	High near ATM / expiry — risk intensifies	
Θ Theta	Always negative (buyers)	Daily time decay ₹	ATM options lose ~₹8–50/day depending on DTE	
ν Vega	Per 1% IV change ₹	Sensitivity to implied volatility	VIX ↑1% → option price ↑ Vega amount	
ρ Rho	Per 1% rate change	Sensitivity to risk-free rate	Small for weeklies; meaningful for LEAPS	

4.2 Greek-Specific Prompts

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Delta hedging NIFTY	I sold 2 lots of NIFTY 23200 CE (delta = 0.35 per lot, lot size 75). What is my total delta exposure in ₹? How many NIFTY futures contracts do I need to buy to delta-hedge this position? NIFTY futures lot size is also 75.	Delta calculation + hedge ratio in ₹ and lots
Gamma risk near expiry	Explain gamma risk for NIFTY options in the last 2 days before weekly expiry. Why does gamma spike for ATM options? If I am short 1 lot of 23000 straddle with spot at 22980 and gamma = 0.005, what happens to my delta if NIFTY moves ₹100?	Gamma risk explanation with NIFTY numbers
Theta decay table	Build a theta decay table for a NIFTY 23000 ATM call with sigma=16.5%, r=5.25%. Show the daily option price and daily theta for DTE = 30, 21, 14, 7, 5, 3, 2, 1. Output as a pandas table.	Theta decay table from 30 DTE to expiry
Vega and India VIX	NIFTY 23000 ATM straddle has vega = 28.5 per lot. India VIX just jumped from 16 to 19 (a 3-point rise). By how much does the straddle premium increase? How does this affect an iron condor position?	Vega impact calculation + effect on condor
Rho sensitivity analysis	RBI cuts repo rate by 25 bps from 5.50%	Rho impact for weekly vs

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
	to 5.25%. What is the impact on NIFTY option prices? Show the rho calculation for a 30-day ATM call and put. Is this impact significant for weekly options?	monthly NIFTY options

05

SECTION

Strategy Builder Prompts

Build, compute and visualise NIFTY F&O strategies

5.1 The 8 Built-in Strategies

The strategy builder includes 8 pre-built strategies. Each automatically computes legs, premiums, lot values, breakevens, and generates a 2D payoff chart + 3D payoff surface. Use these prompts to go deeper.

STRATEGY	OUTLOOK	MAX PROFIT	MAX LOSS	BEST USED WHEN
Long Call	Bullish	Unlimited	Premium paid	Strong directional move expected
Long Put	Bearish	Strike - Premium	Premium paid	NIFTY breakdown / bearish event
Bull Call Spread	Mildly Bullish	Width - Net Debit	Net Debit	Low VIX, limited upside expected
Bear Put Spread	Mildly Bearish	Width - Net Debit	Net Debit	Moderate decline expected
Long Straddle	High Volatility	Unlimited	2x Premiums	Pre-event: Budget/RBI/results
Long Strangle	High Volatility	Unlimited	2x Premiums	Cheap vol; wide move expected
Iron Condor	Neutral	Net Credit	Width - Credit	Low VIX (<18), range-bound NIFTY
Butterfly	Neutral Range	Wing Width - Cost	Net Debit	NIFTY stuck at specific strike

5.2 Strategy-Specific Prompts

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Design Iron Condor	Design a NIFTY weekly iron condor for expiry in 7 days. Spot = 23000, India VIX = 16.5%. Sell the 23400 CE and 22600 PE. Buy the 23600 CE and 22400 PE. Compute: net premium received, max profit, max loss, both breakevens, and probability of profit assuming log-normal distribution.	Full condor P&L table with probability of profit
Size the straddle	I want to buy a NIFTY ATM straddle before the RBI policy announcement. Spot = 23000, K = 23000, DTE = 3, sigma = 22% (elevated pre-event). Compute the straddle cost per lot. What move does NIFTY need to make for me to profit? Show in ₹ and %.	Straddle cost + break-even move calculation
Multi-lot position sizing	I have a capital of ₹5,00,000 for NIFTY options. I want to sell an iron condor collecting ₹8,500 net premium per lot with ₹21,500 margin required per lot.	Position sizing with capital and margin constraints

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
	How many lots can I trade? What is my max portfolio P&L?	
Payoff diagram in Python	Generate a payoff diagram in Python using matplotlib for a NIFTY 23000 long straddle with call premium = ₹145 and put premium = ₹138. Show the profit/loss at expiry for NIFTY ranging from 21000 to 25000. Add breakeven lines. Include lot size of 75.	Python matplotlib payoff diagram with lot-adjusted P&L
Strategy comparison	Compare a NIFTY long straddle vs long strangle for the same capital outlay. Spot=23000, DTE=7, sigma=16.5%. Straddle uses K=23000 for both legs. Strangle uses 23200 CE and 22800 PE. Show P&L table for NIFTY at: 22000, 22500, 23000, 23500, 24000.	Side-by-side comparison table with P&L at each level

06

SECTION

3D Visualization Prompts

Build interactive Plotly surfaces and Three.js backgrounds

6.1 3D Surface Prompts

The model contains 5 interactive 3D surfaces built with Plotly. Use these prompts to build, customise, or add new surfaces.

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Spot × Volatility surface	Build an interactive 3D Plotly surface showing NIFTY call option price across: spot from 21000 to 25000 (step 100) and implied volatility from 8% to 50% (step 2%). Strike K=23000, T=7 days, r=5.25%. Show both call and put surfaces overlaid. Use a dark theme with green for calls and red for puts.	Two overlaid 3D surfaces in Plotly with dark theme
Spot × Time surface	Create a 3D Plotly surface of NIFTY ATM call price vs: spot from 21500 to 24500 and days-to-expiry from 1 to 60. Use K=23000, r=5.25%, sigma=16.5%. Add camera angle and axis labels. Show how time value collapses as DTE approaches 0.	Time decay 3D surface with animated time axis
Volatility smile 3D	Generate a 3D volatility smile chart for NIFTY using Plotly. X axis: strike (21000-25000). Y axis: DTE (7, 14, 30, 60). Z axis: implied volatility with NIFTY skew – OTM puts get 0.3% higher IV per 1% OTM, OTM calls are flat. Use 4 colored lines on a 3D scatter.	NIFTY volatility smile with realistic skew in 3D
Greeks 3D surface	Plot a 3D surface of NIFTY call delta vs spot (21000-25000) and days-to-expiry (1-60). Overlay a second surface for gamma ×100 on the same chart. Use Plotly with transparent surfaces, contour lines, and dark background.	Dual-layer Greeks 3D surface in Plotly
Three.js animated background	Write JavaScript Three.js code for an animated background for my NIFTY options dashboard. I want: a floating wireframe mesh that undulates like an options price surface, particle field in blue/teal, horizontal price-level line segments. Dark background, no shadows. Loop animation at 60fps.	Three.js r128 compatible animated background code

07

SECTION

Live Data Integration Prompts

Yahoo Finance, NSE APIs and real-time data feeds

7.1 Fetching Live Market Data

The tool fetches NIFTY spot price and India VIX from Yahoo Finance (^NSEI, ^INDIAVIX). Use these prompts to work with live data fetching, error handling, and fallbacks.

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Fetch NIFTY from Yahoo Finance	Write JavaScript that fetches the live NIFTY 50 price from Yahoo Finance API endpoint: <code>https://query1.finance.yahoo.com/v8/finance/chart/%5ENSEI?interval=1m&range=1d</code> via the <code>allorigins.win</code> CORS proxy. Extract: current price, day high, day low, previous close, % change. Handle fetch errors gracefully with a fallback to 23000.	Complete JS fetch function with error handling
India VIX fetch	Extend the Yahoo Finance fetch to also retrieve India VIX from ticker ^INDIAVIX. Parse <code>regularMarketPrice</code> from the meta object. If VIX fetch fails, default to 16.5. Show the complete async function.	VIX fetch function with fallback
Python live fetch in Colab	In Google Colab, fetch live NIFTY 50 price using the <code>yfinance</code> library. Get the most recent close price of ^NSEI, the day high/low, and India VIX from ^INDIAVIX. Print them formatted as a quant dashboard summary.	yfinance Python code for NIFTY + VIX in Colab
NSE option chain data	How can I fetch the actual NSE option chain data (not just theoretical BS prices) for NIFTY? What is the NSE API endpoint for option chain? How do I parse the JSON to extract CE/PE prices and OI for each strike?	NSE API endpoint + JSON parsing guide
Auto-refresh logic	Add an auto-refresh mechanism to my NIFTY options HTML page. The page should re-fetch Yahoo Finance data every 3 minutes, update the topbar price, recompute all BS prices, and rebuild the option chain. Show the JavaScript with <code>setInterval</code> and status indicators.	Auto-refresh JS with status bar updates

08

SECTION

Analysis & Interpretation Prompts

Extract insights from BS model outputs

8.1 Market Analysis Prompts

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
IV vs HV comparison	Given NIFTY implied volatility (India VIX / 100) = 16.5% and 30-day historical volatility of NIFTY = 12.8%, compute the IV premium. Is it better to be a buyer or seller of options? How does this affect the iron condor vs straddle choice?	IV/HV analysis with trading recommendation
Probability of expiry range	Using Black-Scholes log-normal assumptions, compute the probability that NIFTY (currently 23000) will expire between 22500 and 23500 in 7 days. Use sigma=16.5%, r=5.25%. Show the formula using N(d2) and provide the Python code.	Probability calculation using N(d2) with Python code
Expected move calculation	What is the 1-standard-deviation expected move for NIFTY over 7 days if India VIX = 16.5? Show the formula: Expected move = $S \times \sigma \times \sqrt{T}$. Convert to percentage and absolute ₹ value. How does this relate to the straddle price?	Expected move in ₹ and % with straddle relationship
Theta harvesting viability	India VIX is at 16.5% (low vol regime). Is this a good environment to sell NIFTY options (theta harvesting)? Show the annualised theta yield of selling an ATM straddle. What VIX level makes option selling unviable?	Theta yield analysis with VIX regime classification
Risk-adjusted returns	I sold a NIFTY iron condor collecting ₹9,200 with margin of ₹22,500 for 7 DTE. Compute: ROI per trade, annualised ROI (assuming 48 weekly trades/year), max risk. Is this risk-adjusted return competitive?	ROI and annualised return calculation for condor

09

SECTION

Advanced & Extension Prompts

Go beyond Black-Scholes — add features to the model

9.1 Model Extensions

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Add dividend yield	Modify the Black-Scholes formula to account for continuous dividend yield q . NIFTY has a dividend yield of approximately 1.2%. How does including q change $d1$ and $d2$? Rewrite the Python BS function with the q parameter. Show the impact on a 30-day call price.	Modified BS formula with dividend yield in Python
Implied volatility solver	Given a NIFTY option market price of ₹182 (call, $K=23200$, $T=7$ days, $S=23000$, $r=5.25\%$), use the Newton-Raphson method to compute the implied volatility. Write the Python iterative solver. What is the implied vol?	Newton-Raphson IV solver in Python with convergence check
Heston model introduction	Explain the Heston stochastic volatility model vs Black-Scholes for NIFTY options. What problem does it solve (volatility smile)? What are the 5 Heston parameters and how are they calibrated to NIFTY option chain data?	Heston model explanation with NIFTY calibration approach
Add Monte Carlo simulation	Add a Monte Carlo simulation to my NIFTY options model. Simulate 10,000 paths of NIFTY spot price over 7 trading days using GBM: $dS = S(r dt + \sigma dW)$. Compute the call and put option prices from the simulation. Compare with BS closed-form prices.	Monte Carlo vs BS comparison in Python
Scenario analysis	Add a scenario analysis table to the NIFTY options model. For a long straddle position, compute P&L for these scenarios: NIFTY up/down 2%, 5%, 10%, and VIX changes to 12%, 16%, 20%, 25%. Show as a 2D heatmap using seaborn.	Scenario matrix with seaborn heatmap

9.2 UI/UX & Dashboard Extension Prompts

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
Add candlestick chart	Add an interactive NIFTY price candlestick chart to the HTML dashboard using Plotly. Fetch historical OHLC data for ^NSEI from Yahoo Finance for the last 30 days. Display with volume bars below. Use the dark theme matching the existing dashboard.	Plotly candlestick + volume chart for NIFTY
Add real-time alerts	Add a price alert system to the NIFTY	Price alert logic with

GOAL / SCENARIO	PROMPT TO USE IN CLAUDE	EXPECTED OUTPUT
	dashboard. If NIFTY moves more than ₹150 from the current price, show a browser notification and flash the topbar red/green. If India VIX crosses 20, show a "HIGH VOL WARNING" banner.	browser notification API
Export to Excel	Add an "Export to Excel" button to the NIFTY option chain. On click, it should download the current option chain data (all strikes, CE/PE prices, Greeks, OI) as a formatted .xlsx file using SheetJS. Include column headers and color-code CE/PE columns.	SheetJS export function for option chain data
Mobile responsive layout	The current NIFTY options dashboard uses a fixed two-column layout (sidebar + content). Make it fully mobile responsive using CSS media queries. On screens <768px, stack the sidebar on top, make the option chain table horizontally scrollable, and reduce font sizes.	CSS media queries for mobile-first responsive layout

10

SECTION

Quick Reference Card

Formulas, NIFTY specs, and prompt patterns

10.1 Black-Scholes Formula Reference

TERM	FORMULA
d1	$d1 = [\ln(S/K) + (r + \sigma^2/2) \cdot T] / (\sigma \cdot \sqrt{T})$
d2	$d2 = d1 - \sigma \cdot \sqrt{T}$
Call C	$C = S \cdot N(d1) - K \cdot e^{-rT} \cdot N(d2)$
Put P	$P = K \cdot e^{-rT} \cdot N(-d2) - S \cdot N(-d1)$
Delta Δ	Call: $N(d1)$ Put: $N(d1) - 1$
Gamma Γ	$N'(d1) / (S \cdot \sigma \cdot \sqrt{T})$
Theta Θ	$[-S \cdot N'(d1) \cdot \sigma / (2 \cdot \sqrt{T}) - r \cdot K \cdot e^{-rT} \cdot N(d2)] / 365$
Vega v	$S \cdot N'(d1) \cdot \sqrt{T} / 100$ (per 1% change in σ)
Rho ρ	$K \cdot T \cdot e^{-rT} \cdot N(d2) / 100$ (per 1% change in r)
Parity	$C - P = S - K \cdot e^{-rT}$ (Put-Call Parity)

10.2 NIFTY F&O Specifications

SPECIFICATION	VALUE	NOTES
Underlying	NIFTY 50 Index	NSE Broad Market Index
Option Type	European (CE/PE)	Cash-settled at expiry
Lot Size	75 units	Changed periodically by SEBI
Strike Interval	₹50	Auto-adjusted near current price
Expiry Cycle	Weekly (Thursday)	Also monthly last-Thursdays
Trading Hours	09:15 – 15:30 IST	Monday to Friday
Risk-Free Rate	5.25% (RBI Repo)	As of March 2026 post cut
India VIX Range	10 – 40+ typical	<15 low, 15–25 normal, >25 high
Settlement	Cash (INR)	No physical delivery
Margin	SPAN + Exposure	NSE SPAN margin model

10.3 Prompt-Writing Best Practices for Claude

TIP 1 — ALWAYS SET CONTEXT FIRST

Begin every session with: "Act as a quantitative finance researcher building a NIFTY 50 options pricing dashboard in Python/JavaScript." This primes Claude to use Indian market conventions, ₹ currency, and NSE lot sizes.

TIP 2 — SPECIFY EXACT NIFTY PARAMETERS

Always include: spot price (e.g., 23000), strike (e.g., 23000), DTE (e.g., 7 days), $r=5.25\%$ (RBI repo), $\sigma=\text{India VIX}/100$ (e.g., 16.5%), lot size=75. Vague prompts give generic results.

TIP 3 — ASK FOR NIFTY-SPECIFIC CONTEXT

Add "in the context of NSE weekly options" or "using NIFTY lot size of 75" to ensure Claude tailors explanations to Indian markets rather than generic US equity options (lot=100, USD pricing).

TIP 4 — REQUEST FORMATTED OUTPUT

For code: "Format as a Google Colab cell with markdown headings." For analysis: "Output as a pandas DataFrame." For explanations: "Use numbered steps with a worked example at NIFTY=23000."

TIP 5 — ITERATE WITH FOLLOW-UPS

After getting base code, ask: "Now add error handling for when Yahoo Finance is unreachable," or "Now convert the Python function to JavaScript for a browser-based slider." Claude maintains context across the conversation.

NIFTY LIVE OPTIONS LAB

Complete Prompt Guide for Claude AI

Black-Scholes Model · NSE India · 10 Sections · 50+ Prompts
Option Chain · Strategy Builder · 3D Surfaces · Live Yahoo Finance Data
[NSEQuant Lab](#) · [March 2026](#) · [claude.ai](#)

FREE PROMPT GUIDE ·

AI MARKET DIRECTION PROBABILITY MODEL PROMPT GUIDE

By

@depthinksfinance

How to Build a 3D Quant Finance Models Using Claude AI

Logistic Regression · Random Forest · Gradient Boosting · Plotly 3D Charts

3

ML MODELS

5

COPY PROMPTS

3D

3D CHARTS

FREE

TOOLS NEEDED

STEP 00 **WHAT YOU WILL BUILD**

The final output you'll have after following this guide

After following this 5-step guide, you will have a fully interactive 3D financial dashboard that predicts market direction using real machine learning — the exact same model shown in the viral post.

Example Output · NIFTY 50 Prediction

60.9% PROBABILITY UP	39.1% PROBABILITY DOWN	+1.32% EXPECTED MOVE	70.7% CONFIDENCE
--------------------------------	----------------------------------	--------------------------------	----------------------------

What the Dashboard Includes

- 3 ML models running simultaneously: Logistic Regression, Random Forest, Gradient Boosting
- Interactive 3D charts: rotate, zoom, and explore probability surfaces and scatter clouds
- Feature importance analysis showing what drives market direction
- Rolling walk-forward accuracy test across 18 years
- Confusion matrices, correlation heatmap, and probability timeline
- Dark Bloomberg-style theme with animated ticker tape

Tool Required

Claude.ai (free tier works) — go to claude.ai · Click the paperclip icon to upload your data file · Then paste the prompts below one by one. No coding knowledge needed.

STEP 01 **PREPARE YOUR DATA**

What file format and columns you need

You need a spreadsheet with monthly or weekly historical prices of your market index (NIFTY, SENSEX, S&P 500, etc.) plus related global variables. The more variables, the smarter the model.

COLUMN NAME	WHAT IT IS	REQUIRED?	EXAMPLE
MONTH / DATE	Monthly date (first of each month)	YES — Required	2008-01-01
NIFTY / INDEX	Your target index closing price	YES — Required	25,285
S&P 500	US large cap index	Recommended	4,769
DOW JONES	US blue chip index	Recommended	37,689
GOLD	Gold spot price (USD/oz)	Recommended	2,063
BRENT CRUDE OIL	Crude oil price (USD/bbl)	Recommended	77.04
USD INR	Dollar to Rupee exchange rate	Optional	83.10
US 10 YR BOND	US Treasury yield (%)	Optional	3.88%
INDIA VIX	Implied volatility index	Optional	14.5
MSCI EM	Emerging market index	Optional	1,034

Minimum Requirement

At least 5 years of monthly data (60 rows) to train meaningful ML models. More data = better predictions. The ideal is 10-18 years. Excel .xlsx or CSV both work.

STEP 02 THE MASTER PROMPT

The single prompt that builds everything — paste this into Claude after uploading your file

This is the core prompt. It builds the complete ML pipeline, all three models, feature engineering, cross-validation, and the full 3D HTML dashboard in a single response. Upload your Excel/CSV first, then paste this.

PROMPT 1 — MASTER PROMPT (COPY & PASTE INTO CLAUDE)

You are a quantitative finance analyst and Python ML developer.
I have attached a dataset with historical monthly price data of [YOUR INDEX].
The dataset includes columns like: Date, Close Price, and multiple market variables (global indices, commodities, bonds, currencies, etc.).

Your task: Build a complete AI Directional Probability Model that predicts whether the market will go UP or DOWN next month.

STEP 1 - DATA PREPARATION:

- Load the dataset and identify the primary index column
- Sort chronologically, handle missing values
- Calculate monthly log returns for all columns

STEP 2 - FEATURE ENGINEERING (create these features):

- Lagged returns: lag-1, lag-2, lag-3 of the main index
- Lagged returns of: global indices, gold, oil, DXY, FX rate
- Moving average deviations: price vs 3M and 6M moving average
- Momentum: 3-month and 6-month price momentum
- Rolling volatility: 3-month standard deviation of returns
- Yield spread: domestic 10Y bond minus US 10Y bond

STEP 3 - TARGET VARIABLE:

- Binary: 1 if next month return > 0 (UP), else 0 (DOWN)

STEP 4 - TRAIN 3 ML MODELS (Time-Series 5-Fold Cross Validation):

1. Logistic Regression (L2 regularization, C=0.5)
2. Random Forest (200 trees, max_depth=5, min_samples_leaf=8)
3. Gradient Boosting (150 estimators, max_depth=3, lr=0.05)

STEP 5 - PREDICTIONS for current market conditions:

- Probability of UP (%) for each model
- Probability of DOWN (%) for each model
- Weighted ensemble (weighted by CV accuracy)
- Expected move (probability-weighted historical avg return)
- Confidence = how far ensemble is from 50/50

STEP 6 - EVALUATION:

- Walk-forward rolling accuracy (12-month smoothed window)
- Confusion matrix for all 3 models
- Feature importance across all models
- Feature correlation heatmap

STEP 7 - GENERATE A COMPLETE HTML DASHBOARD with:

- Hero section with ensemble prediction (big numbers)
- Individual model cards with probability bars
- 3D Monte Carlo paths chart (drag-to-rotate)
- 3D feature importance surface (models x features)
- 3D probability evolution surface over time
- 3D scatter cloud (LR x RF x GB, colored by actual outcome)
- Rolling accuracy chart, confusion matrices, radar chart
- Dark Bloomberg-style theme, Plotly.js, animated ticker
- ALL data embedded as JSON inside the HTML file

OUTPUT: Single self-contained HTML file, opens in any browser.

Pro Tip

After uploading your file, mention your index column name explicitly — e.g. 'The main index is NIFTY 50 in the column called NIFTY' so Claude identifies it correctly without guessing.

STEP 03 ADD LIVE MARKET DATA

Update predictions with today's real prices for maximum accuracy

Use this follow-up prompt in the same chat after Prompt 1. Fill in today's actual market values before sending — Claude will update the prediction numbers with fresh live data.

PROMPT 2 — LIVE DATA UPDATE

Now update the model with today's live market data:

Current date: [TODAY'S DATE — e.g. March 10, 2026]

Current NIFTY 50 price: [e.g. 24,158]

India VIX: [e.g. 23.36] ← find at nseindia.com

USD/INR rate: [e.g. 92.30]

Brent Crude Oil: [e.g. 74.50 USD/bbl]

Gold price: [e.g. 2,910 USD/oz]

S&P 500 current level: [e.g. 5,614]

NIFTY month-to-date return: [e.g. -6.2%]

Instructions:

- Use these values as the current feature vector for prediction
- For volatility: blended vol = 70% implied (VIX/100) + 30% historical
- Update the dashboard hero section with fresh predictions
- Add a "Live Data as of [DATE]" badge on the dashboard
- Search the web for any missing current data points needed

Where to Find Live Data

NSE India (nseindia.com) for India VIX · Google Finance / Moneycontrol for index prices · RBI website for Indian bond yields · Investing.com for oil, gold & global indices

STEP 04 BONUS — MONTE CARLO*Add GBM price path simulation to the dashboard*

This optional prompt adds a Monte Carlo simulation layer on top of the directional model. It simulates 10,000 possible price paths and gives you a full range of outcomes — not just direction but price targets.

PROMPT 3 — MONTE CARLO ADDON

Add a Monte Carlo simulation section to the dashboard:

Parameters:

- Model: Geometric Brownian Motion (GBM)
- Simulations: 10,000 paths
- Time horizon: 30 trading days
- Starting price: [CURRENT INDEX PRICE]
- Daily drift (mu): estimate from historical log returns
- Daily volatility (sigma): 70% VIX implied + 30% historical

GBM Formula:

$Price(t+1) = Price(t) \times \exp((\mu - 0.5 \times \sigma^2) \times dt + \sigma \times \sqrt{dt} \times Z)$
 where Z is a standard normal random variable

Calculate and display:

- Expected price after 30 days (mean of all simulations)
- Median price (P50)
- P5 downside risk / P95 upside potential
- P(price up), P(price down)
- P(+5% gain), P(-5% loss)

Add 3D charts:

1. 3D Monte Carlo paths (500 sampled paths)
2. 3D probability density wave (day x price x density)
3. Confidence bands (P5/P25/mean/P75/P95)
4. Final price distribution histogram

Show a combined ML + Monte Carlo signal at the top:

If ML says UP 60% AND Monte Carlo shows P(up) 55%,
 display: **STRONG BULL SIGNAL** with combined confidence score.

STEP 05 **CUSTOMIZATION PROMPTS**

Personalize the dashboard for your audience

Prompt 4A — Change Your Index

CUSTOMIZE INDEX NAME

Replace all references to "NIFTY 50" with [YOUR INDEX NAME].
Update the dashboard title, ticker tape, and all labels.
The index I am using is: [e.g. SENSEX / BANK NIFTY / S&P 500 / Nikkei 225]
Current price: [CURRENT PRICE]
Trading currency: [INR / USD / EUR / JPY]

Prompt 4B — Change Dashboard Theme

CUSTOMIZE VISUAL THEME

Change the dashboard color theme to one of these options:

- "Bloomberg orange-black"
- "Robinhood green-white"
- "TradingView dark-blue"
- "Gold luxury dark"
- "Matrix green on black"
- "Clean white minimal"

Keep all charts and data identical. Only change:
background colors, accent colors, font choices, card borders.

Prompt 5 — Social Media Summary Card

CREATE SHAREABLE CARD

Create a summary card for sharing on social media (1080x1080px):
Show only: Index name, current price, UP probability, DOWN probability,
expected move %, confidence %, 3 model signals side by side.
Include a clean bar chart comparing the 3 models.
Add my watermark/handle: [YOUR SOCIAL HANDLE]
Style: bold, clean, easy to read as a screenshot.
Output as a self-contained HTML file.

HOW TO USE · 6-STEP WORKFLOW

STEP	ACTION	DETAILS
01	Go to claude.ai	Open Claude in your browser. Free account works fine. Click New Chat.
02	Upload your data file	Click the paperclip icon. Upload your Excel (.xlsx) or CSV file. Wait for it to fully upload before typing.
03	Paste Prompt 1	Copy the Master Prompt. Replace [YOUR INDEX] with your actual index name. Send it. Claude takes 1-3 minutes.
04	Download the HTML file	Click download on the code block Claude generates, or copy the HTML and save as dashboard.html. Open in Chrome.
05	Add live data (Prompt 2)	In the same chat, paste Prompt 2 with today's prices filled in. Get an updated HTML with fresh predictions.
06	Share your dashboard	Screenshot for social media, or host free on netlify.app (drag and drop). The HTML works offline too.

Understanding Your Results

METRIC	WHAT IT MEANS	WHAT IS GOOD
P(UP) / P(DOWN)	Ensemble probability the market closes higher/lower next month.	Above 60% = meaningful signal. Near 50% = uncertain.
Expected Move %	Probability-weighted average return based on historical UP/DOWN months.	Any non-zero value gives you a directional bias.
Confidence %	How far the ensemble is from a 50/50 coin flip. Max 100% = certain.	Above 20% = worth noting. Above 40% = high conviction.
CV Accuracy	Cross-validation accuracy on out-of-sample historical periods.	52-58% is realistic. 50% is random baseline.
Feature Importance	Which variables drive the model most. Higher bar = more influence.	Variables with importance above 10% are key signals.

Honest Reality Check

Real-world ML accuracy on monthly market direction is typically 52-58%. That is only slightly above random (50%). These models identify structural tendencies — they are NOT a crystal ball. Always use as one signal among many inputs, never as a sole trading signal.

Troubleshooting Common Issues

PROBLEM	LIKELY CAUSE	FIX
Claude can't find index column	Column name is unclear	Add to prompt: The main index is in column called [exact column name]
Model accuracy is exactly 50%	Too little data (under 60 rows)	Get more historical data. 10+ years monthly is ideal.
3D charts don't rotate	Old browser / CDN blocked	Use Google Chrome. Check internet connection for Plotly CDN.
HTML output is cut short	Claude hit output length limit	Send: The HTML was cut off. Please continue from where you stopped.
Predictions seem incorrect	Data gaps or bad values	Add to prompt: Handle missing values by forward-filling or dropping rows.
Want different index	Default is set to NIFTY	Use Customization Prompt 4A to change the index name throughout.

Level Up — Advanced Ideas

- **Add Sentiment Analysis:** Tell Claude: Also add a sentiment score based on FII vs DII flow ratio. Include this as a feature in the models.
- **Add Technical Indicators:** Tell Claude: Add RSI (14-period), MACD signal, Bollinger Band position, and ADX trend strength as additional features.
- **Multi-Index Comparison:** Tell Claude: Run the same model for NIFTY, BANKNIFTY, and SENSEX. Show which has the highest confidence signal this month.
- **Weekly Prediction Model:** Tell Claude: Convert this to a weekly model using daily closing prices to predict the next trading week.
- **Regime Detection:** Tell Claude: Add a regime classifier (Bull/Bear/Sideways) based on rolling 3M return + volatility. Show accuracy per regime.

If you found this useful, share it with your network.

The more people learn quantitative finance, the better our ecosystem gets.

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*This is an educational prompt guide only. Not financial advice. ML models cannot reliably predict markets. All investments carry risk.
Past performance does not guarantee future results.*